Phytomedicine 98 (2022) 153978

Contents lists available at ScienceDirect

Phytomedicine

journal homepage: www.elsevier.com/locate/phymed

Original Article

Scientific basis for medicinal use of *Citrullus lanatus* (Thunb.) in diarrhea and asthma: *In vitro, in vivo* and *in silico* studies

Muqeet Wahid, Fatima Saqib *

Department of Pharmacology, Faculty of Pharmacy, Bahauddin Zakariya University, Multan (60000), Pakistan

ARTICLE INFO

Keywords:
Citrullus lanatus
Watermelon
LC ESI-MS/MS
Bronchodilator
Asthma
Antidiarrheal

ABSTRACT

Background: Citrullus lanatus (Thunb.) is a member of the Cucurbitaceae family, commonly farmed as an edible vegetable around the globe. It has been used in traditional therapies in addition to nutritional advantages. Traditional herbal practitioners employ *C. lanatus* seeds to treat gastrointestinal, respiratory, and urinary diseases in Pakistan and India. However, more investigation is needed to understand the effect of *C. lanatus* seeds on treating gastrointestinal, respiratory, and urinary disorders.

Purpose: This research aimed to use network pharmacology and molecular docking to understand multi-target mechanisms of *C. lanatus* seeds against asthma and diarrhea and to validate its effects using biological tests to investigate antispasmodic and bronchodilator capabilities.

Methods: The ground seeds of *C. lanatus* were extracted in hexane, dichloromethane, ethanol, and aqueous for sequential extracts. The bioactive components in sequential extracts of *C. lanatus* seeds were identified using LC ESI-MS/MS, and specific compounds were quantified using HPLC. The quantified bioactive compounds of *C. lanatus* were subjected to *in silico* studies for network pharmacology and molecular docking to elucidate their role in antispasmodic and bronchodilator properties. The sequential extracts were tested on isolated rabbit tissue, i.e., jejunum, trachea, and urinary bladder. The antiperistalsis, antidiarrheal and antisecretory studies were also performed in animal models.

Results: In silico studies indicate that bioactive chemicals from sequential extracts of *C. lanatus* seeds interfere with asthma and diarrhea-associated pathogenic genes. Those are members of calcium mediate signaling, cholinergic synapse, regulation of cytosolic calcium concentration, smooth muscle contraction, and inflammatory responses. It was also found that rutin, quercitrin, stearic acid, umbelliferone, and kaempferol were stronger binding to voltage-gated calcium channels and muscarinic M3 receptor, thus exerting calcium channel blocker activity and cholinergic receptor stimulant response. On isolated jejunum, trachea, and urinary preparations, sequential extracts of *C. lanatus* seeds elicited the spasmolytic response and showed the relaxation of spastic contractions of K^+ (80 mM) and carbachol (1 μM). Furthermore, it induced a non-parallel rightward shift in calcium concentration-response curves with suppression. In animal models, *C. lanatus* seed extracts exhibited partially or completely antiperistalsis, antidiarrheal, and antisecretory effects.

Conclusion: Thus, *Citrullus lanatus* had therapeutic benefits by modulating the contractile response through calcium-mediated signaling target proteins, hence exerting bronchodilator and antidiarrheal properties. The current study provides evidence for further mechanistic studies and the development of *C. lanatus* seeds as a potential therapeutic intervention for patients with gastrointestinal, respiratory, and urinary disorders.

Abbreviation

CAMK2B calcium/calmodulin-dependent protein kinase IIB
C.C.B calcium channel blocker
CCh carbachol
Cl.Aqueous aqueous extract
Cl. D.C.M. dichloromethane extract
Cl.Ethanol ethanol extract
Cl *C. lanatus* seeds
Cl.Hexane *n*-hexane extract

Abbreviations: ACN, acetonitrile.

* Corresponding author.
E-mail address: fatima.saqib@bzu.edu.pk (F. Saqib).

https://doi.org/10.1016/j.phymed.2022.153978
Received 17 May 2021; Received in revised form 10 January 2022; Accepted 2 February 2022
Available online 4 February 2022

C.R.C. concentration-response curves
G.O. gene ontology
KEGG kyoto encyclopedia of genes and genomes
Log Ki logarithmic of inhibition constant (K_i)
MM-GBSA molecular mechanics energies combined generalized born and surface area
MLCK-1 myosin light chain kinase-1
TFA trifluoroacetic acid
VGCC voltage-gated calcium channel β2a

Introduction

The scientific data revealed the expediency of phototherapeutic agents to treat chronic diseases as an adjunct or alternative therapy. If a dietary pattern contains a larger volume of vegetables, fruits, and grains, ingestion plays a crucial role in preventing chronic diseases due to non-nutrient secondary metabolites (Patel and Rauf, 2017; Wahid et al., 2020). *Citrullus lanatus* (Thunb.) Matsum. & Nakai. (**Common name:** Watermelon; **Folk name**: Tarbooz or Tarbuz, **Ayurvedic:** Kalinga, **Unani:** Tarbuz, **Seed:** Tukhm-e-Tarbooz) (Khare, 2007) is the member Cucurbitaceae family widely cultivated as edible fruit or vegetable in the entire world about 6.8% of the vegetable farming devoted area occupied by *C. lanatus* production (Renner and Pandey, 2013). Due to the highest nutritional value of *C. lanatus*, the pink flesh is consumed fresh, canned, or processed. Commonly, *C. lanatus*'s flesh is eaten raw and roasted. However, it also consumed desserts, salad, snacks, watermelon cake, lemonade, and watermelon rind pickles (Erhirhie and Ekene, 2014). In summer, drinks and juices from *C. lanatus* flesh and seeds are made for freshness, chilling, and thirst-quenching (Erhirhie and Ekene, 2014). The seeds and fruit of *C. lanatus* as traditional medicine for urinary complaints, hepatic congestion, intestinal catarrh, and gastrointestinal disorders (Khare, 2007). Studies show that *C. lanatus* seeds and fruit have analgesic, anti-inflammatory, diuretic, and anti-urolithiatic (Siddiqui et al., 2018) antioxidant (Gill et al., 2011), hypotensive, cardioprotective (Massa et al., 2016), laxative (Sharma et al., 2012), and anti-ulcerative activities (Gill et al., 2011).

Despite phytoconstituents and nutritional constituents, *C. lanatus* seed was dumped as industrial waste. However, proximate and phytochemical analyzes revealed that *C. lanatus* seed might have health advantages (Perkins-Veazie et al., 2007; Rabiu and Muhammad, 2015; Reetu and Tomar, 2017). *C. lanatus* seed rich in proteins (32.1%), fats (59.2%), fiber (8.2%), carbohydrates (4.5%) (Alka et al., 2012; El-Adawy and Taha, 2001; Morais et al., 2017). *C. lanatus* seeds contain numerous amino acid citrulline with smaller L-arginine, leucine, isoleucine, histidine, lysine phenylalanine, methionine, valine threonine, methionine, serine, glutamic acid, and aspartic acid (Taiwo et al., 2009). Mabaleha et al. (2007) and Taiwo et al. (2009) reported that *C. lanatus* seeds are rich in linoleic acid, oleic acid, stearic acid, and palmitic acid (Ouassor et al., 2020; Raihana et al., 2015; Taiwo et al., 2009). Beside fatty acid, it contains phytosterols, including sitosterol, stigmasterol, and stigmastadienol (Ouassor et al., 2020). Abu-Reidah et al. (2013) reported polar fraction of *C. lanatus* fruit and seed is richer in polyphenols and other polar phytoconstituents. *C. lanatus* contains L-citrulline, β-Sitosterol, protocatechuic acid 3-glucoside, kaempferol-3-O-glucorhamnoside, 5-O-caffeoyl shikimic acid, p-coumaric acid glucoside I, 6-O-feruloylsucrose, 3-feruloylquinic acid, rutin, dihydrokaempferol 7-glucoside, calodendroside A, narcissin (isorhamnetin 3-O-rutinoside), isoorientin, orientin, isovitexin, ajugol, quercitrin, coumarin, and saligenin glucopyranoside, (Abu-Reidah et al., 2013; Siddiqui et al., 2018).

Thus, the main objective of this study was to (1) prepare bioactive extracts of *C. lanatus* seeds, (2) characterize bioactive compounds in extracts using chromatography, (3) evaluate antidiarrheal and bronchodilator properties, and (4) investigate pharmacodynamic mechanisms of *C. lanatus* seeds using *in vitro, in vivo, and in silico* models.

Materials and methods

Preparation of extract

From May to July 2018, *C. lanatus* seeds were obtained from fresh fruits pulp farmed in local farms in Multan, Punjab, Pakistan, rendering them free of fruit pulp. A taxonomist Dr. Zafarullah Zafar, from Institute of Pure and Applied Biology, Bahauddin Zakariya University Multan, Pakistan, identified (herbarium voucher number: P.K.F. 1916/30) the seeds and fruit of *C. lanatus*. A plant specimen was submitted in the same institute's Herbarium with the voucher number P.K.F. 1916/30.

After removing the husks from the seed, the interior section was crushed into coarse seeds powder using an herbal grinder. The pulverized material (120 g) was hot macerated in the Soxhlet apparatus for sequential extraction (fractions) based on solvent polarity. The solvents were evaporated in a rotating evaporator at their boiling temperatures under reduced pressure and maintained at 32 ± 5 °C to yield oils (n-hexane and dichloromethane) and thick brownish yellowish tinted extracts (ethanolic and aqueous). The overall yield of *C. lanatus* seeds extracts was 59.1%, whereas n-hexane, DCM, ethanol, and aqueous yields were 32 g, 17 g, 14 g, and 8 g, respectively. The bioactive extracts were transferred to an amber container and maintained at a temperature of −20 °C. On the experiment day, nonpolar n-hexane and DCM extracts were prepared in 1.5% v/v Tween-80, whereas the polar ethanol and aqueous extracts were prepared in distilled water or 0.9% normal saline diluted to the appropriate concentration.

Chemicals

Analytical grade solvents and chemicals used in this study were procured from *Merck & Co.*, Darmstadt, Germany, and m *Sigma Chemicals*, Co., St. Louis, MO, USA.

Sample preparation for HPLC and LC-ESI-MS/MS

The sequential polarity-based *C. lanatus* seed extracts were diluted in 1 ml 100% methanol for HPLC and LC-ESI-MS/MS analysis, and the supernatant was collected after centrifugation for 12 min at 14,000 rpm. The supernatant was filtered using a 0.22 μm syringe filter.

LC ESI-MS/MS analysis

LC ESI-MS/MS (LTQ XL mass spectrometer, Thermo Electron Corporation, Waltham, MA, USA) was used to analyze polarity-based sequential extracts of *C. lanatus* seeds to conduct a preliminary screening of bioactive compounds (Seraglio et al., 2016; Wahid et al., 2021). The direct injection of sample into an electron spray ionization (ESI) probe configured to both positive ($M+$) and negative (M-) ionization modes. The mobile phase used to separate ethanol and aqueous extracts compounds consisted of solvent A: 0.1% formic acid in methanol and Solvent B: 0.1% formic acid in water and acetonitrile (80:20). Similarly, the mobile phase employed for the separation of compounds in the *n*-hexane and DCM extracts consisted of solvent A: 0.1% formic acid in methanol and Solvent B: 0.1% formic acid in acetonitrile. The optimal chromatographic parameters for separation were; mobile phase flow rate (0.4 ml/min), the column temperature (25 °C), sample injection volume (8 μL), mass-spectrum range (m/z = 50–1000), and column size (C-18, 2.1 mm × 100 mm, 3 μm). The mobile phase gradient was adjusted for solvent A (v/v) as follows: 98%, 98–80%, 80–70% 70–10% and 10–98% from 0–4.0 min, 4.0–8.0 min, 8.0–15.0 min, 15.0–17.0 min, and 17.0–20.0 min, respectively. The ideal ion source settings for mass spectrometric analysis were; ion spray voltage for positive mode: 5500 V, ion spray voltage for negative mode: −5500 V; curtain gas (CUR): 25 psi; nebulizer gas (GS1): 55 psi; auxiliary gas (GS2): 55 psi; ion source temperature: 400 °C. As a CUR and GS2, nitrogen was employed. Ions are isolated in an ion trap and fragmented using collision-induced dissociation (CID) energies ranging from 10 to 45, depending on the stability of the parent precursor ions used in tandem mass spectrometry. Thermo Scientific's X-Calibur 3.0 (Waltham, MA, USA) was used to gather and process ESI-MS/MS data.

Identification of compounds

The bioactive constituents were identified by comparing their mass spectra chromatogram to previously published data and reference libraries, including the Mass Bank of North America (Mo-NA; http://mona.fiehnlab.ucdavis.edu, Accessed on April 8, 2021) and the European Mass Bank (https://massbank.eu/MassBank/, Accessed on April 8, 2021). The spectra of compounds were compared to high-resolution masses and MS/MS fragmentation of libraries and databases (Syed et al., 2021). ChemDraw 18.0 (PerkinElmer informatics, Waltham, MA, USA) was used to elucidate the structure.

Quantification of bioactive compounds by using analytical HPLC-DAD UV/Vis

RP-HPLC method optimization and validation

The polarity-based DCM, ethanol, and aqueous extracts of *C. lanatus* seeds were processed to RP-HPLC (Agilent Technologies, Germany) to validate and quantify the bioactive compounds (epicatechin, ferulic acid, rutin, caffeic acid, quercetin, umbelliferone, 1,4-dicaffeoylquinic acid, quinic acid, kaempferol, stigmasterol, apigenin, scopoletin, malic acid, p-coumaric acid, apigenin-7-O-glucuronide, luteolin 7-O-glucoside, and hesperidin) (Qamar et al., 2021; Wahid et al., 2021). The sample was analyzed in RP-HPLC using a C-18 column (C18, 4.6 × 150 mm, 5 μm), and the following chromatographic parameters were optimized; mobile phase, column temperature, injection volume, flow rate, and detection wavelengths to visualize peaks. We employed a binary eluent system in which the mobile phase consists of solvent A: 0.1% TFA in methanol and Solvent B: 0.1% TFA in the water and ACN (80:20) to separate compounds of the ethanol and aqueous extracts. In contrast, we have used a mobile phase comprised of solvent A: 0.1% TFA in methanol and solvent B: 0.1% TFA in ACN to separate compounds in DCM extract. The optimal chromatographic conditions were; temperature (25 °C), mobile phase flow rate (0.8 ml/min), column sample injection volume (8 μL), and optimal detection wavelengths (250, 270, 280, and 320 λ). The mobile phase gradient was adjusted for solvent A (v/v) as follows: 95%, 95–90%, 90–80% 80–10% and 10–95% from 0–4.0 min, 4.0–8.0 min, 8.0–14.0 min, 14.0–20.0 min, and 20.0–30.0 min, respectively. The results were compared with UV spectra and the retention time (Rt) of external standards.

Validation of an analytical method

The validation approach was established in accordance with the International Conference on Harmonization (ICH) principles and detailed in a previous study (Wahid et al., 2021).

Preparation of external standards: A stock solution of external standards (epicatechin, ferulic acid, rutin, caffeic acid, quercetin, umbelliferone, 1,4-dicaffeoylquinic acid, quinic acid, kaempferol, stigmasterol, apigenin, scopoletin, malic acid, p-coumaric acid, apigenin-7-O-glucuronide, luteolin 7-O-glucoside, and hesperidin) dissolved in 1 ml 100% methanol at concentration 150 μg/ml and the supernatant was collected after centrifugation for 10 min at 14,000 rpm; stored at −20 °C. On the analysis day, the stock solution was diluted to further 100 and 50 μg/ml. The supernatant was collected and filtered using a 0.22 μm syringe filter.

Linearity, limits of detection, and quantification: For linearity validation, dilutions (7.81–500 μg/ml) of external standards were prepared to assess the linearity detector response by constructing a standard curve based on the concentration and the peak area of each standard. The detection limits (LOD) and quantification limits (LOQ) were calculated using the formula below.

$$\mathrm{LOD} = 3.3\sigma \div \mathrm{S}$$

$$\mathrm{LOQ} = 10\sigma \div \mathrm{S}$$

where σ is the intercept standard deviation and S is the slope of the linear regression equation.

Specificity: A total volume of 8 μl of extracts and external standards samples were injected in RP-HPLC under the optimal chromatographic parameters mentioned above. The chromatographic peaks and retention time (Rt) of extracts and external standards were compared to determine peak purity. The calibration curves for each external standard were used to quantify the peaks.

Precision and repeatability: The precision and repeatability of the current method for each external standard were studied on interday ($n = 4$) and intraday ($n = 4$ on three consecutive days) under the optimal chromatographic parameters mentioned above, and the relative standard deviation (RSD) were determined.

Accuracy and recovery: The accuracy and recovery of the current method for each external standard were determined by retrieving external standards from sequential extracts of the *C. lanatus* seeds samples. Three different strengths of external standards (50, 100, and 150 μg/ml) were used for this method. 50, 100, and 150 μg of external standards and 100 mg of polarity-based sequential extracts of *C. lanatus* seeds samples were mixed to get a final concentration of external standards; 50, 100, and 150 μg/ml. A total volume of 8 μl mixture sample was injected in RP-HPLC under the optimal chromatographic parameters mentioned above. The sample recovery percentage of an external standard was estimated using the following equation.

$$\%Recovery = (observedconcentration \div actualconcentration) \times 100$$

Animals and housing conditions

The protocols were approved by the Department of Pharmacology ethical committee (vide No. EC /04-PhDL/S2018), and the studies strongly aligned with *Commission of Laboratory Animal Resources* (Rowan, 1979). Sprague-Dawley rats (170 ± 20 g) and mice (28 ± 3 g) were obtained without discrimination of gender for experimental purposes from the same faculty. The animals were housed in the animal house that followed a dark/light cycle with standard food and free access to tap water under controlled circumstances (22 ± 4 °C). Rabbits were slaughtered with a sharp knife for *in vitro* experiments, whereas rats and mice were killed through cervical dislocation for *in vivo* experiments.

In silico approaches

Network pharmacology analysis

The methodology described by Xiao et al. (2021) was used for network pharmacology.

Screening potential targets: The potential protein targets of bioactive compounds (epicatechin, ferulic acid, rutin, caffeic acid, quercetin, umbelliferone, 1,4-dicaffeoylquinic acid, quinic acid, kaempferol, stigmasterol, apigenin, scopoletin, malic acid, p-coumaric acid, apigenin-7-O-glucuronide, luteolin 7-O-glucoside, and hesperidin) were obtained from Drugbank (https://go.drugbank.com, Accessed date: 28th April 2021) and Swiss Target Prediction (http://www.swisstargetprediction.ch, Accessed date: 28th April 2021) databases. The Gene card (https://www.genecards.org, Accessed date: 28th April 2021), DisGeNET (http://www.disgenet.org/web/DisGeNET, Accessed date: 28th April 2021), Pubmed (https://pubmed.ncbi.nlm.nih.gov, Accessed date: 28th April 2021), and Online Mendelian Inheritance in Man (OMIM) (https://www.omim.org, Accessed date: 28th April 2021) databases were searched for target genes associated with gastrointestinal and tracheal disorders using the keywords "diarrhea"; "constipation"; "irritable bowel syndrome"; "asthma"; "coughing". The VarElect gene card tool (https://ve.genecards.org/, Accessed date: 28th April 2021) was used to examine the targets of gastrointestinal and tracheal illnesses for clinical phenotype and genetic association and sorted the top 150 disease target genes. The intersection analysis was conducted between bioactive compounds and disease target genes in R studio in conjunction with the package "ggvenn" and Venn Diagram was built to demonstrate the correlation between bioactive compound and disease target genes. KEGG pathway and gene ontology (GO) enrichment were investigated further for corelative targets.

Construction of network and pathway analyzes: R studio v1.4 was used in conjunction with the packages "BiocManager", "Clusterprofiler", and "DOSE" using database "org.Hs.eg.db" and following parameters: "Homo sapiens", and p-value and q-value cutoff must be equal to 0.05 for KEGG pathway and gene ontology (GO) enrichment. The compound target disease proteins (C-T-D) network, the Protein-Protein Interaction (PPI) network, and the Compounds-Targets-Pathways (C-T-P) network were all constructed and analyzed in Cytoscape 3.8.0.

Molecular docking

The methodology described by Sirous et al. (2019) was used for Molecular docking experiments.

Ligand Preparation: The prominent bioactive compounds in network pharmacology were used as ligands further to investigate binding forces and stability in target proteins. The three-dimensional structures of ligands were obtained from PubChem (https://pubchem.ncbi.nlm.nih.gov, Accessed date: 30th April 2021) in mol2 format and subjected in LigPrep (Maestro v11.8, Schrodinger suite 2018–4) for ligand ionization, minimization, and optimization. OPLS3e force field was applied for energy minimization and optimization of ligands to enhance accuracy and performance of predicting protein-ligand model that provides the lowest energy conformer. Following Epik tool was used for possible protomers, and ionization states were generated at target pH (7.4 ± 2.0). The tautomeric states were generated for each ionized ligand and used the Desalt to retain the ligands with the largest number of atoms. The stereoizer was used to generate stereoisomers of each ligand and retained stereogenic chirality data from the input file, with a limit of 10 stereoisomers considered per ligand.

Protein Preparation: The three-dimensional crystallized structure of proteins was obtained from RCSB Protein Databank (RCSB PDB) (https://www.rcsb.org, Accessed date: 28th April 2021) in PDB format and subjected to Protein Preparation module (Maestro v11.8, Schrodinger suite 2018–4) for protein generating het states, hydrogen bond assignment, ionization, minimization, and optimization. The imported protein structure was determined whether the protein-ligand complex is dimer or multimer and refined the structure to remove redundant binding site. Herein, bond orders were adjusted for untemplated amino acid residues and het group based on the chemical component dictionary (CCD) database. The covalent bond from metal ions and nearby atoms to protein structure changed to zero-order and corrected the formal charges to metal ions and nearby atoms. If the protein structure contains sulfurs close to each other within 3.2 Å, then disulfide bonds were created between two sulfurs. The Prime tool was used to fill missing side chains atoms and loops in protein structure. Finally, in preprocess Epik tool is used to generate the protonation and metal charge states of the het groups at target pH (7.4 ± 2.0) for ligands. Then all water molecules and Het groups from the protein structure were deleted beyond 5.00 Å, and hydrogen atoms were added to the structure. PROPKA tool was used to optimize the H-bond assignment of protein structure at pH 7.0, and the OPLS3e force field was used to constrain energy minimization for energy reduction and protein structure optimization.

Molecular Docking and Receptor grid generation: With receptor grid generation (Maestro v11.8, Schrodinger suite 2018-4), the binding pockets of the protein structure were defined for molecular docking. A literature review, a selection of previously bound protein ligands or sitemap (Maestro v11.8, Schrodinger suite 2018-4) were used to define coordinates (x,y,z) of the cubic grid box. The following coordinates were used for 1T0J (x: −3.348, y: 3.044, z: −12.198), 5ZHP (x: 2.21, y: −43.17, z: 237.18), 3BHH (x: 21.92, y: 31.1, z: 11.18), and 6C6M (x: −31.678, y: 19.497, z: 12.316). The grid box's length has been increased to 16 Å pixels. The potential of the receptor's nonpolar portions was reduced by a factor of 1.0 Å based on the Van-der Waals radii of nonpolar atoms of protein with a partial charge cut-off of 0.25 Å

The extra precision (XP) Glide (Maestro v11.8, Schrodinger suite 2018-4) was used for induced fit molecular docking of prepared ligands and proteins and utilizing a previously generated receptor grid file. The partial charge was set at a cut-off of 0.15 Å and the scaling factor of 0.80 Å for Vander Waals radii. The ligand sampling was adjusted to flexible docking and included the nitrogen (non-ring) inversions and ring conformations to generate conformers. The bias sampling of the torsions around the bond was also included. The Epik states penalties were added to the docking score.

The XP docking findings were exposed to Prime MM-GBSA (Maestro v11.8, Schrodinger suite 2018-4), which used the VSGB solvation with the force field OPLS3e to determine the binding energies of ligands with protein structures.

Inhibition Constant (Ki): According to the equation below, the inhibition constant was calculated using the binding free energy of the ligand

$$\Delta G = -RT(lnKi) or\ Ki = e(-\Delta G / RT)$$

Where ΔG is binding free energy of ligand, R is gas constant (cal.mol^{-1}. K^{-1}), and T is room temperature (298 K).

Isolated tissue experimentation

The methodology described by Saqib and Janbaz (2016) and Wahid et al. (2021) was used for *in vitro* experiments.

Isolated rabbit jejunum preparations

The jejunum tissue of a rabbit was dissected. The surrounding mesenteries were excised from tissue, and 2 to 3 cm long jejunum tissue segments were prepared. Each jejunal preparation was fixed in a 15 mL tissue organ bath that had been previously filled with the Tyrode's solution accreted with carbogen and maintained at 37 °C using a circulating thermoregulator. Before testing the drug, a preloaded force of 1 ± 0.1 g was applied to jejunal preparation. Next, jejunal preparation was allowed to equilibrate for 25 ± 5 min with buffer drainage with new Tyrode's solution; after 8 ± 2 min, spontaneous rhythmic contractions were recorded. An isotonic transducer (MLT0015) was used to record tissue physiological responses, which were amplified by a data acquisition device Power Lab® (4/25) and visualized in Lab Chart Pro (Version 7). The percentage of contraction obtained immediately before the test drug dosage was used to determine the physiological response. To investigate the spasmolytic and spasmogenic effect of polarity-based n-hexane (Cl.Hexane), DCM (Cl.DCM), ethanol (Cl.Et), and aqueous (Cl.Aq) extracts of *C. lanatus* seeds, the jejunal preparation was materialized in a tissue organ bath (Janbaz et al., 2015).

The antispasmodic effect of polarity-based sequential extracts of *C. lanatus* seeds was assessed by eliciting smooth muscle contractions by blocking calcium ion channels with K^+ (80 mM) or opening potassium ion channels with K^+ (25 mM). The extracts were applied cumulatively to obtain a concentration-dependent inhibitory response of polarity-based sequential extracts of C. lanatus seeds. K^+ (80 mM) may influence smooth muscle contraction by causing a calcium ion influx into the cell, therefore depolarizing it. The extract was referred a calcium channel blocker due to its ability to prevent this K^+ (80 mM) induced contraction (Saqib and Janbaz, 2021).

To determine the calcium antagonistic response of polarity-based sequential extracts of *C. lanatus* seeds, the jejunal preparation was treated three times with K^+ (80 mM) to reduce intracellular calcium concentration and then equilibrated for 20 ± 5 min in a calcium free Tyrode solution containing EDTA, then filled with calcium free and potassium rich Tyrode's solution and equilibrated for 40 ± 5 min. Calcium was added to the jejunum preparation cumulatively after 20 ± 5 min of incubation to produce calcium concentration response curves (CRCs). A persistent increase in the contractile-response of tissue illustrates that extracellular calcium ions are required for smooth muscle contractions to be reliable. When control CRCs were produced, washed the jejunal preparation and incubated for 50 ± 10 min with polarity-based sequential extracts of *C. lanatus* seeds, calcium CRCs were then formed. These CRCs were produced at various concentrations of polarity-based sequential extracts of *C. lanatus* seeds and compared to control CRCs to detect probable calcium antagonism (Saqib and Janbaz, 2021).

Isolated rat ileum preparation

Rat ileum tissue was dissected, surrounding mesenteries were excised, and 2 to 3 cm long ileum tissue segments were prepared. Each ileum preparation was fixed in a 15 mL tissue organ bath that had been previously filled with the Tyrode's solution accreted with carbogen and maintained at 37 °C using a circulating thermoregulator. A preloaded force of 1 ± 0.1 g was applied to ileum preparation and allowed to equilibrate for 30 ± 5 min with draining the buffer to replace it with fresh after every 10 ± 2 min; quiescent contractions were observed. The isolated rat ileum was incubated in a tissue organ bath to assess the cholinergic response to n-hexane and dichloromethane extracts of *C. lanatus* seeds (Gilani et al., 2008). To test the contractile response on a resting baseline, bolus doses of n-hexane and dichloromethane extracts of C. *lanatus* seeds were induced into rat ileum preparations.

Isolated rabbit tracheal preparations

The tracheal tissue of a rabbit was dissected. The surrounding fatty and other adhesion materials were excised from tissue, and 3 to 4 mm width rings were prepared. The tracheal rings were incised laterally to produce a tracheal strip; as a consequence, smooth muscles were interposed between cartilaginous segments of the strip. Each tracheal preparation was fixed in a 15 mL tissue organ bath that had been previously filled with the Kreb's solution accreted with carbogen and maintained at 37 °C using a circulating thermoregulator. Before testing the drug, a preloaded force of 1 ± 0.1 g was applied to tracheal preparation. Next, tracheal preparation was allowed to equilibrate for 40 ± 5 min with buffer drainage with new Kreb's solution; after 8 ± 2 min. An isometric transducer (FORT100) was used to record tissue physiological responses, which were amplified by a data acquisition device Power Lab® (4/25) and visualized in Lab Chart Pro (Version 7). The percentage of contraction obtained immediately before the test drug dosage was used to determine the physiological response. To investigate the bronchodilator effect of polarity-based n-hexane (Cl.Hexane), DCM (Cl. DCM), ethanol (Cl.Et), and aqueous (Cl.Aq) extracts of *C. lanatus* seeds, the tracheal preparation was materialized in a tissue organ bath (Janbaz et al., 2015).

The tracheal preparation was exposed to K^+ (80 mM), carbachol (1 μM), and K^+ (25 mM) elicited contractions to investigate the putative bronchodilator activity of polarity-based sequential extracts of *C. lanatus* seeds. The extracts were applied cumulatively to obtain a concentration-dependent inhibitory response of polarity-based sequential extracts of *C. lanatus* seeds. The carbachol CRCs were formed in the presence and absence of polarity-based sequential extracts of *C. lanatus* seeds. Carbachol was added cumulatively to form carbachol control CRCs (three times), and tissue was rinsed with Kreb's solution to reconstruct the baseline tension. When control CRCs were formed, tracheal preparations were incubated for 50 ± 10 min with polarity-based sequential extracts of *C. lanatus* seeds; carbachol CRCs were then formed. These CRCs were formed using polarity-based sequential extracts of *C. lanatus* seeds at various concentrations and compared to control CRCs to investigate putative muscarinic receptor antagonism.

Isolated urinary bladder preparations

The urinary bladder tissue of a rabbit was dissected. The urinary bladder rings were incised laterally to produce a urinary bladder strip; as a consequence, smooth muscles were interposed between cartilaginous segments of the strip. Each urinary bladder preparation was fixed in a 15 mL tissue organ bath that had been previously filled with the Kreb's solution accreted with carbogen and maintained at 37 °C using a circulating thermoregulator. Before testing the drug, a preloaded force of 1 ± 0.1 g was applied to urinary bladder preparation. Next, urinary bladder preparation was allowed to equilibrate for 40 ± 5 min with buffer drainage with new Kreb's solution; after 8 ± 2 min. An isometric transducer (FORT100) was used to record tissue physiological responses, which were amplified by a data acquisition device Power Lab® (4/25) and visualized in Lab Chart Pro (Version 7). The percentage of contraction obtained immediately before the test drug dosage was used to determine the physiological response. To investigate the bronchodilator effect of polarity-based n-hexane (Cl.Hexane), DCM (Cl.DCM), ethanol (Cl.Et), and aqueous (Cl.Aq) extracts of *C. lanatus* seeds, the urinary bladder preparation was materialized in a tissue organ bath (Janbaz et al., 2015).

The urinary bladder preparation was exposed to K^+ (80 mM), carbachol (1 μM), and K^+ (25 mM) elicited contractions to investigate the putative bronchodilator activity of polarity-based sequential extracts of *C. lanatus* seeds. The calcium CRCs were formed in the presence and absence of polarity-based sequential extracts of *C. lanatus* seeds for calcium antagonistic activity. To determine the calcium antagonistic response of polarity-based sequential extracts of *C. lanatus* seeds, the urinary bladder preparation was treated three times with K^+ (80 mM) to reduce intracellular calcium concentration and then equilibrated for 20 ± 5 min in a calcium free Tyrode solution containing EDTA, then filled with calcium free and potassium rich Tyrode's solution and equilibrated for 40 ± 5 min. Calcium was added to the jejunum preparation cumulatively after 20 ± 5 min of incubation to produce calcium concentration response curves (CRCs). A persistent increase in the contractile-response of tissue illustrates that extracellular calcium ions are required for smooth muscle contractions to be reliable. When control CRCs were produced, washed the urinary bladder preparation and incubated for 50 ± 10 min with polarity-based sequential extracts of *C. lanatus* seeds; calcium CRCs were then formed. These CRCs were produced at various concentrations of polarity-based sequential extracts of *C. lanatus* seeds and compared to control CRCs to detect probable calcium antagonism (Saqib and Janbaz, 2021).

In vivo experimentation

Evaluation of maximum tolerated dose

In order to determine the maximum tolerated dose of *C. lanatus* polarity-based sequential extracts, six groups of rats were selected for each extract. Normal saline and five dosages of polarity-based sequential *C. lanatus* extract at 50, 100, 150, 200, and 300 mg/kg per day were administered for 28 days. Body weight, clinical indications of distress, behavioral changes, and animal mortality were observed over 28 days. (Elasoru et al., 2021).

Charcoal meal GI transit test

The methodology described by Saqib and Janbaz (2016) was used for gastrointestinal transit. The food was withdrawn from mice overnight before the study and housed in separate cages. Each animal was assigned to a group; each group had five animals.

Group I: as a negative control, orally received 0.9% normal saline (10 ml/kg)
Group II: as a positive control, orally received normal saline (10 ml/kg p.o.) and charcoal meal (10 ml/kg) after 15 min
Group III: loperamide group, orally received 10 mg/kg of loperamide
Group IV: verapamil group, orally received 10 mg/kg of verapamil

Group V and VI: orally received 150 and 300 mg/kg of hexane extract of *C. lanatus* seed
Group VII and VIII: orally received 150 and 300 mg/kg of DCM extract of *C. lanatus* seed
Group IX and X: orally received 150 and 300 mg/kg of ethanol extract of *C. lanatus* seed
Group XI and XII: orally received 150 and 300 mg/kg of aqueous extract of *C. lanatus* seed

On trial, doses of polarity-based sequential extracts of *C. lanatus* seeds (150 and 300 mg/kg) were chosen and given to Group V to XII.

After 15 min of treatment, all groups except Group I received an oral dose of the charcoal meal (10 mL/kg) comprising 20% starch, 10% vegetable charcoal, and 10% gum acacia in normal saline. After thirty minutes, mice were slaughtered by cervical dislocation, and their abdomens were opened to remove the whole small intestine to determine the distance traveled by the charcoal meal. The percent of the peristaltic index was determined as $D_t/L_i \times 100$, where D_t is the distance traveled by charcoal meal and L_i is the length of an animal's small intestine.

Castor oil-induced diarrhea

The methodology described by Wahid et al. (2021) was used for Castor oil-induced diarrhea. The food was withdrawn from mice overnight before the study and housed in separate cages. Each animal was assigned to a group; each group had five animals.

Group I: as a negative control, orally received 0.9% normal saline (10 ml/kg)
Group II: as a positive control, orally received normal saline (10 ml/kg p.o.) and castor (10 ml/kg p.o.) after 30 min
Group III: loperamide group, orally received 10 mg/kg of loperamide
Group IV: verapamil group, orally received 10 mg/kg of verapamil
Group V and VI: orally received 150 and 300 mg/kg of hexane extract of *C. lanatus* seed
Group VII and VIII: orally received 150 and 300 mg/kg of DCM extract of *C. lanatus* seed
Group IX and X: orally received 150 and 300 mg/kg of ethanol extract of *C. lanatus* seed
Group XI and XII: orally received 150 and 300 mg/kg of aqueous extract of *C. lanatus* seed

On trial, doses of polarity-based sequential extracts of *C. lanatus* seeds (150 and 300 mg/kg) were chosen and given to Group V to XII.

After 30 min of treatment, all groups except Group I received an oral dose of castor oil (10 ml/kg). Six hours later, wet diarrhea patches on white paper were observed in the cages, and the mean defecation per group was computed. The percentage inhibition was calculated as $(D_c - D_t)/D_c \times 100$, where D_c represents the mean defecation of the control group and D_t represents the mean defecation of the test group.

Castor oil induce intestinal fluid accumulation

The methodology described by Wahid et al. (2021) was used for Castor oil-induced intestinal fluid accumulation that was modified for this study. The food was withdrawn from mice overnight before the study and housed in separate cages. Each animal was assigned to a group; each group had five animals.

Group I: as a negative control, orally received 0.9% normal saline (10 ml/kg)
Group II: as a positive control, orally received normal saline (10 ml/kg p.o.) and castor (10 ml/kg p.o.) after 30 min
Group III: loperamide group, orally received 10 mg/kg of loperamide
Group IV: verapamil group, orally received 10 mg/kg of verapamil
Group V and VI: orally received 150 and 300 mg/kg of hexane extract of *C. lanatus* seed
Group VII and VIII: orally received 150 and 300 mg/kg of DCM extract of *C. lanatus* seed
Group IX and X: orally received 150 and 300 mg/kg of ethanol extract of *C. lanatus* seed
Group XI and XII: orally received 150 and 300 mg/kg of aqueous extract of *C. lanatus* seed

On trial, doses of polarity-based sequential extracts of *C. lanatus* seeds (150 and 300 mg/kg) were chosen and given to Group V to XII.

After 30 min of treatment, all groups except Group I received an oral dose of castor oil (10 ml/kg). After thirty minutes, mice were slaughtered by cervical dislocation, and their abdomens were opened for the small intestine. The small intestine was ligated at both ends, pylorus and caecum, and the whole small intestine was removed. The small intestine was weighed with and without intestinal fluid, and the results were computed using the formula (Pi/Pm)1000, where Pm is the mice's body weight, and Pi is the intestine's weight. Intestinal fluid accumulation data were given as weight of fluid (g).

Statistical analysis

The results of the experiments were represented as mean ± standard deviation (S.D.), and sigmoidal dose-response graphs with a nonlinear regression curve fit were used to calculate the median effective concentration (EC_{50}) with a 95% confidence interval (CI). Concentration-response curves were drawn using logarithmic sigmoidal dose-response graphs. For *in vivo* research, one-way ANOVA was used, followed by the Dunnett test, with a $p < 0.05$ deemed significant. Graphpad Prism (Version 8.0) was used for all statistical analyzes and graph charting.

Results

Identification of bioactive compounds by LC-ESI-MS/MS analysis

LC ESI-MS/MS analysis identified 42 bioactive compounds in *C. lanatus* sequential extracts. The phytochemical analysis of sequential extracts of *C. lanatus* seeds using LC ESI-MS/MS indicated the presence of sterols, anthraquinones, phenols, flavones, and flavonoids. Table 1 and Fig. 1 illustrate the essential bioactive phytoconstituents that may together be responsible for this biological activity. The hexane extract contains linoleic acid (ESI MS fragment ions (m/z): 279.25, 261.25, 259.17, and 243.17 $[M\text{-}H]^-$), palmitoleic acid (ESI MS fragment ions (m/z): 253.21, and 217.8 $[M\text{-}H]^-$), oleic acid (ESI MS fragment ions (m/z): 281.25, 263.25, 261.17, 237.17, and 207.17 $[M\text{-}H]^-$), stearic acid (ESI MS fragment ions (m/z): 283.33, 265.17, and 239.17 $[M\text{-}H]^-$), and campesterol (ESI MS fragment ions (m/z): 383.4, and 161 $[M\text{-}H_2O+H]^+$). The DCM extract contains umbelliferone (ESI MS fragment ions (m/z): 161, 133, 117 $[M\text{-}H]^-$), caffeic acid (ESI MS fragment ions (m/z): 179.17, 172, 161.17, and 135 $[M\text{-}H]^-$), stigmasterol (ESI MS fragment ions (m/z): 397, 379, 355, 342.1, and 297 $[M + H]^+$) and D-mannitol (ESI MS fragment ions (m/z): 59, 71,73, 89, 101, 113, 119, 163, and 181 $[M\text{-}H]^-$). The ethanol extract contains scopoletin (ESI MS fragment ions (m/z): 192, 191, 176, 148, 137, and 104 $[M\text{-}H]^-$), quinic acid (ESI MS fragment ions (m/z): 191,173, 171, 127, 93, and 85 $[M\text{-}H]^-$), epicatechin (ESI MS fragment ions (m/z): 290, 289, 245, 179, 165.08, and 151.25 $[M\text{-}H]^-$), ferulic acid (ESI MS fragment ions (m/z): 181, 164, 133, 118, 114.1, and 104 $[M\text{-}H]^-$), rutin (ESI MS fragment ions (m/z): 609.25, 301.08, and 271 $[M\text{-}H]^-$), quercetin (ESI MS fragment ions (m/z): 301.08, 286, 273.08, 179, 151, 121, and 107 $[M\text{-}H]^-$), apigenin (ESI MS fragment ions (m/z): 269, 240, 225, 151, and 117 $[M\text{-}H]^-$), and 1,4-dicaffeoylquinic acid (ESI MS fragment ions (m/z): 471, 394, 353, 341 327.25, 317, 299.25, 234, 216, and 203 $[M\text{-}H]^-$). The aqueous extract contains apigenin-7-O-glucuronide (ESI MS fragment ions (m/z): 445, 269, 175, and 151 $[M\text{-}H]^-$), luteolin 7-O-glucoside (ESI MS fragment ions (m/z): 285.08, 284.08, 256.25, 227.08 211.08,

Table 1
Identification of bioactive compounds in sequential extracts of *Citrullus lanatus* (Thunb.) seeds corresponding to the chromatographic fragments by LC ESI-MS/MS.

Sequential Extracts	Rt (min)	Molecular Weight	Observed MS (m/z)	Calculated MS (m/z)	Error (ppm)	Precursor type	ESI-IT MS/MS (Ions)	Empiricalformula	Proposed compound	Class
Hexane	0.96	278.4	279.232	279.2315	−1.8	[M-H]-	279.25, 261.25, 259.17, 243.17	$C_{18}H_{30}O_2$	Linoleic acid	Polyunsaturated fatty acid; omega-6
	0.97	254.41	253.2159	253.2161	0.8	[M-H]-	253.21, 217.8	$C_{16}H_{30}O_2$	Palmitoleic acid	Fatty acid
	1.02	282.5	281.2478	281.2483	1.8	[M-H]-	281.25, 263.25, 261.17, 237.17, 207.17	$C_{18}H_{34}O_2$	Oleic acid	Fatty acid
	1.08	284.5	283.2643	283.264	−1.1	[M-H]-	283.33, 265.17, 239.17	$C_{18}H_{36}O_2$	Stearic acid	Fatty acid
	3.86	400.68	383.4012	383.4009	−0.8	[M-H2O+H]+	383.4, 161	$C_{28}H_{48}O$	Campesterol	Sterol; Ergosterols and derivatives
	6.2	256.42	255.2322	255.2318	−1.6	[M-H]-	255, 235, 217, 181	$C_{16}H_{32}O_2$	Palmitic acid	Fatty acid
	7.2	242.4	243.2319	243.2329	4.1	[M + H]+	242.21, 241.21, 223.20, 92.7	$C_{15}H_{30}O_2$	Pentadecanoic acid	Fatty acid
	22.6	430.71	431.3886	431.3883	−0.7	[M + H]+	432, 430, 205, 166, 165, 136	$C_{29}H_{50}O_2$	α-tocopherol	Fat-soluble vitamin
DCM	1.3	126.11	127.039	127.0385	−3.9	[M + H]+	95, 81.08, 79,	$C_6H_6O_3$	Maltol	
	1.62	162.14	161.0246	161.0249	1.9	[M-H]-	161, 133, 117,	$C_9H_6O_3$	Umbelliferonee	Hydroxycoumarins
	1.65	173.21	172.0971	172.0973	1.2	[M-H]-	130, 172	$C_8H_{15}NO_3$	N-Acetyl-leucine	Amino acids and derivatives
	1.78	180.16	179.0349	179.0347	−1.1	[M-H]-	179.17, 172, 161.17, 135,	$C_9H_8O_4$	Caffeic acid	Hydroxycinnamic acids
	3.1	166.17	165.0557	165.0556	−0.6	[M-H]-	129, 147, 149, 165	$C_9H_{10}O_3$	Caffeyl alcohol	Cinnamyl alcohols
	3.84	412.69	413.8479	413.8482	0.7	[M + H]+	397, 379, 355, 342.1, 297,	$C_{29}H_{48}O$	Stigmasterol	Sterol
	4.8	131.13	130.0501	130.0499	−1.5	[M-H]-	130.1, 112, 84, 82.3, 66	$C_5H_9NO_3$	cis-4-Hydroxyproline	Amino acids and derivatives
	6.5	182.17	181.0711	181.0712	0.6	[M-H]-	59, 71,73, 89, 101, 113, 119, 163, 181	$C_6H_{14}O_6$	D-Mannitol	Sugar alcohols
	7.4	196.16	195.0503	195.0499	−2.1	[M-H]-	75,99, 101,129, 159, 177	$C_6H_{12}O_7$	Gluconic acid	Sugar acids and derivatives
Ethanol	4.3	178.18	383.4012	383.4009	−0.8	[M-H]-	177, 135, 133, 119	$C_{10}H_{10}O_3$	4-Methoxycinnamic acid	Cinnamic acids
	4.63	192.17	191.0367	191.0365	−1	[M-H]-	192, 191, 176, 148, 137, 104	$C_{10}H_8O_4$	Scopoletin	Hydroxycoumarins
	5.58	192.17	191.0573	191.0576	1.6	[M-H]-	191,173, 171, 127, 93, 85	$C_7H_{12}O_6$	Quinic acid	carboxylic acid
	6.45	290.27	289.0717	289.0719	0.7	[M-H]-	290, 289, 245, 179, 165.08, 151.25	$C_{15}H_{14}O_6$	Epicatechin	Catechin/ Flavonoids
	9.85	472.7	471.348	471.3476	−0.8	[M-H]-	471, 425, 405, 363, 221	$C_{30}H_{48}O_4$	Corosolic acid	Triterpenoids
	9.95	194.18	193.0509	193.0512	1.6	[M-H]-	193, 178, 149, 134	$C_{10}H_{10}O_4$	Isoferulic acid	Cinnamic acids
	10.05	194.18	193.0505	193.0503	−1	[M-H]-	181, 164, 133, 118, 114.1, 104	$C_{10}H_{10}O_4$	Ferulic acid	Cinnamic acids

(*continued on next page*)

Table 1 (*continued*)

Sequential Extracts	Rt (min)	Molecular Weight	Observed MS (*m/z*)	Calculated MS (*m/z*)	Error (ppm)	Precursor type	ESI-IT MS/MS (Ions)	Empiricalformula	Proposed compound	Class
	12.25	610.5	609.1466	609.1461	−0.8	[M-H]-	609.25, 301.08, 271	$C_{27}H_{30}O_{16}$	Rutin	Flavonoid glycosides
	15.51	302.23	301.0359	301.0354	−1.7	[M-H]-	301.08, 286, 273.08, 179, 151, 121,107	$C_{15}H_{10}O_7$	Quercetin	Flavonoid
	16.8	270.24	269.045	269.0455	1.9	[M-H]-	269, 240, 225, 151, 117	$C_{15}H_{10}O_5$	Apigenin	Flavonoids
	17.5	516.4	515.013	515.0128	−0.4	[M-H]-	471, 394, 353, 341 327.25, 317, 299.25, 234,216, 203	$C_{25}H_{24}O_{12}$	1,4-Dicaffeoylquinic acid	Quinic acids
Aqueous	0.75	504.437	503.1617	503.1615	−1.8	[M-H]-	485, 341, 221, 217 179, 161	$C_{18}H_{32}O_{16}$	Maltotriose	Oligosaccharides
	1.4	153.14	152.0339	152.0334	−0.7	[M-H]-	153, 152, 108, 107, 91	$C_7H_7NO_3$	3-Hydroxyanthranilic acid	Hydroxybenzoic acid derivatives
	3.15	4.34	445.0776	445.0779	4.1	[M-H]-	445, 269, 175, 151	$C_{21}H_{18}O_{11}$	Apigenin-7-O-glucuronide	Flavonoid glucuronides
	3.2	448.37	447.0933	447.093	−0.7	[M-H]-	285.08, 284.08, 256.25, 227.08 211.08, 151.17	$C_{21}H_{20}O_{11}$	Luteolin 7-O-glucoside	Flavonoid glycosides
	4.8	434.3	433.0773	433.0771	−1.6	[M-H]-	415, 389, 301,	$C_{20}H_{18}O_{11}$	Quercetin-3-Arabinoside or Quercetin-3-O-xyloside	Flavonoid glycosides
	5.5	172.18	503.1617	503.1615	−1.5	[M-H]-	171, 127, 114, 98	$C_7H_{12}N_2O_3$	Glycyl-l-proline	Amino acids derivatives
	6.5	155.15	154.0603	154.0601	−3.9	[M-H]-	137, 110, 93, 81, 80, 67	$C_6H_9N_3O_2$	l-Histidine	Amino acids derivatives
	7.3	134.09	133.0129	133.0128	1.8	[M-H]-	132, 115, 71	$C_4H_6O_5$	Malic acid	Carboxylic acid
	7.6	164.16	163.0392	163.0391	0.8	[M-H]-	163.0, 119, 93, 65	$C_9H_8O_3$	p-coumaric acid	Hydroxycinnamic acids
	7.94	147.13	146.044	146.0444	−1.1	[M-H]-	146, 128, 102, 61	$C_5H_9NO_4$	Glutamic acid	Amino acids derivatives
	8.7	610.6	609.1817	609.1826	1.5	[M-H]-	609, 301, 285, 151	$C_{28}H_{34}O_{15}$	Hesperidin	Flavonoid glycosides
	16.7	286.23	285.0405	285.0406	−0.8	[M-H]-	286, 285, 215, 151, 142.97, 117	$C_{15}H_{10}O_6$	Kaempferol	Flavanols

and151.17 [M-H]$^-$), malic acid (ESI MS fragment ions (m/z): 132, 115, and 71 [M-H]$^-$), p-coumaric acid (ESI MS fragment ions (m/z): 163.0, 119, 93, and 65 [M-H]$^-$), hesperidin (ESI MS fragment ions (m/z): 609, 301, 285, and 151 [M-H]$^-$), and kaempferol (ESI MS fragment ions (m/z): 286, 285, 215, 151, 142.97, and 117 [M-H]$^-$).

Optimization of HPLC conditions and method validation

The HPLC parameters were optimized with a mobile phase system consisting of solvent A (0.1% TFA in methanol) and solvent B (0.1% TFA in aqueous and acetonitrile) or solvent A (0.1% TFA in methanol) and solvent B (0.1% TFA in acetonitrile)at a flow rate (0.8 ml/min) was used to achieve the maximum separations and peak visibility of compounds over a 50 ± 5 min run time.

To enhance the visibility of chromatographic spikes, the retention times of separation peaks were compared to the retention times of external standards under identical chromatographic circumstances at various wavelengths (λ) 250, 270, 280, and 320 nm (Table 2 and Fig. 2). At 270 nm, umbelliferone, stigmasterol, and caffeic acid constituents of DCM extract were visible. Quinic acid, rutin, and ferulic acid appeared as visible peaks at 280 nm, while epicatechin, 1,4-dicaffeoylquinic acid, scopoletin, apigenin, and quercetin and appeared as visible peaks at 320 nm. The apigenin-7-O-glucuronide and hesperidin constituents of the aqueous extract had visible peaks at 250 nm, while p-coumaric acid, malic acid, kaempferol, and luteolin 7-O-glucoside had peaks at 280 nm.

For linearity validation, the calibration curves of dilutions of external standards were employed. Epicatechin, ferulic acid, rutin, caffeic acid, quercetin, umbelliferone, 1,4-dicaffeoylquinic acid, quinic acid, kaempferol, stigmasterol, apigenin, scopoletin, malic acid, p-coumaric acid, apigenin-7-O-glucuronide, luteolin 7-O-glucoside, and

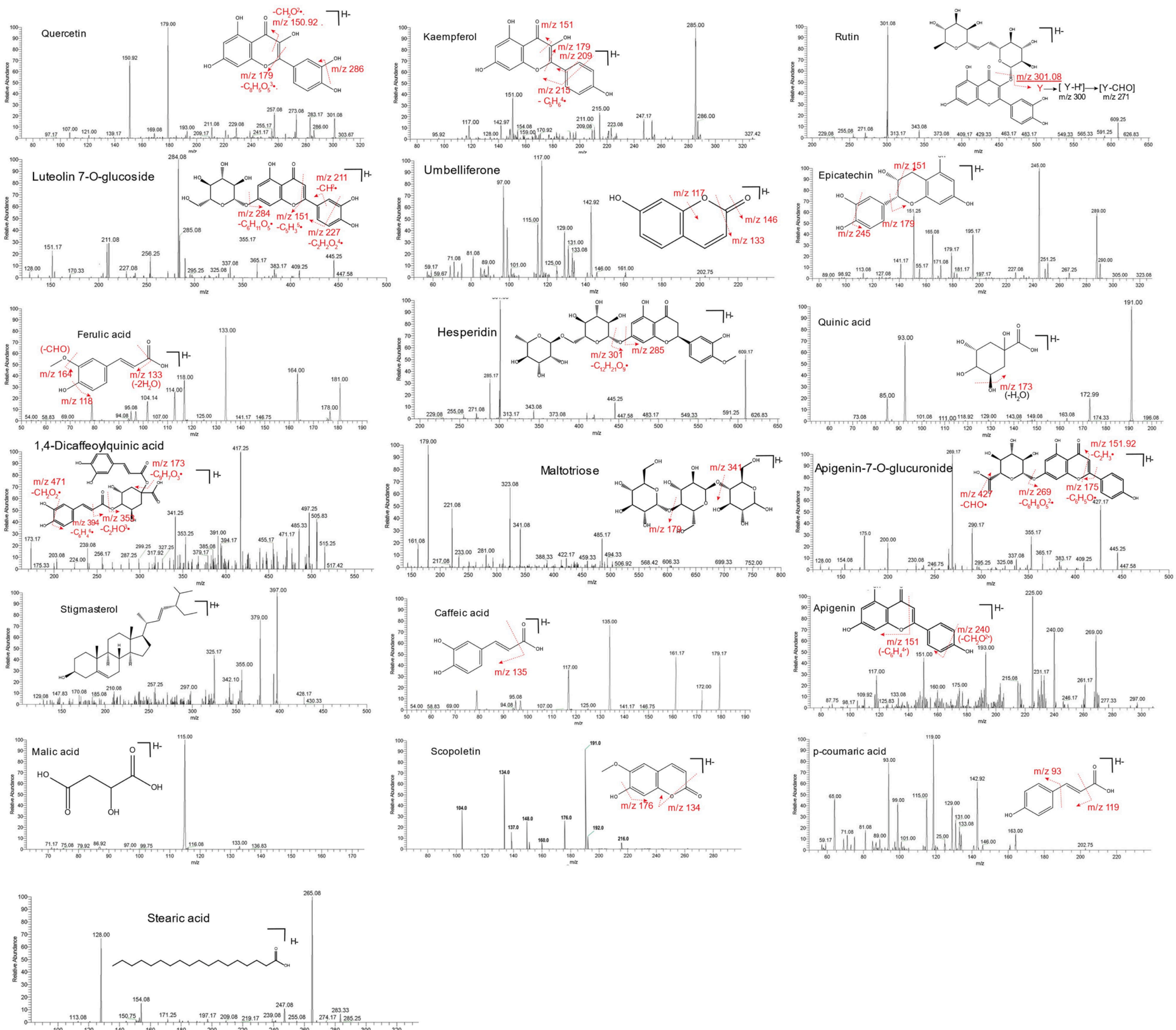


Fig. 1. ESI-MS/MS spectra in negative and positive mode for tentative compounds of sequential extracts of *C. lanatus* seeds. The tentative compounds are quercetin, kaempferol, rutin, luteolin 7-O-glycoside, umbelliferone, epicatechin, ferulic acid, hesperidin, quinic acid, 1,4-dicaffeoylquinic acid, maltotriose, apigenin-7-O-glucuronide, stigmasterol, caffeic acid, apigenin, malic acid, scopoletin, p-coumaric acid, and stearic acid.

hesperidin had a linearity range of 7.81–500 μg/ml. The LOD and LOQ of the present method were determined to be between 0.32–1.10 μg/ml and 0.76–3.34 μg/ml, respectively, as indicated in Table 2. Interday ($n = 4$) and intraday ($n = 12$) evaluation of external standards of epicatechin, ferulic acid, rutin, caffeic acid, quercetin, umbelliferone, 1,4-dicaffeoylquinic acid, quinic acid, kaempferol, stigmasterol, apigenin, scopoletin, malic acid, p-coumaric acid, apigenin-7-O-glucuronide, luteolin 7-O-glucoside, and hesperidin were used to confirm the instrumental precision and repeatability. The precision validation results in Table 2 and Table S1 showed that the interday and intraday% RSDs were 0.43–1.29% and 0.48–1.41%, respectively. To assess the recovery percentage, *C. lanatus* polarity-based sequential extracts were spiked with various concentrations of external standards (50, 100, 150 μg/ml). The percentage recovery findings in Tables 2 and S2 indicated that it varied from 98.44 ± 0.70 to 99.55 ± 0.95, with a% RSD of 0.50–1.34%, indicating that this procedure is accurate.

Quantification analysis of bioactive compounds by HPLC

LC ESI-MS/MS identified bioactive compounds were quantified based on standard calibration curves of external standard dilutions. Table 2 presented the concentration of bioactive compounds in sequential extracts, rutin with higher concentration 788.64 μg/g, followed by kaempferol (683.09 μg/g), quercetin (566.94 μg/g), hesperidin (492.37 μg/g), 1,4-dicaffeoylquinic acid (446.78 μg/g), apigenin-7-O-glucuronide (395.3 μg/g), apigenin (383.49 μg/g), umbelliferone (374.22 μg/g), p-coumaric acid (371.31 μg/g), luteolin 7-O-glucoside (349.81 μg/g), caffeic acid (284.72 μg/g), stigmasterol (260.02 μg/g), ferulic acid (248.4 μg/g), scopoletin (225.89 μg/g), epicatechin (170.46 μg/g), quinic acid (127.68 μg/g), malic acid (104.05 μg/g) were depicted in Table 2. Overall, rutin and kaempferol had higher proportions than other bioactive compounds.

Table 2
Quantification and method validation of bioactive compounds of sequential extracts of *Citrullus lanatus* (Thunb.) seeds.

Sequential Fractions	Analytes	Λ(nm)	Rt (min)	Linear Regression Data			LOD(µg/ml)	LOQ(µg/ml)	Concentration (µg/g)	Precision (RSD%)		Recovery		Analytes + Extract (µg/g)	
				Range (µg/ml)	Equation	r^2				Inter Day	Intra Day	Mean	RSD %	50 µg	100 µg
DCM	Umbelliferone	270	7.3	7.81–500	$y = 29.914x + 4.33$	0.9999	0.57	1.72	374.22	1.16	0.88	99.39 ± 0.73	0.74	420.82	471.69
	Stigmasterol		18.2	7.81–500	$y = 33.589x + 5.16$	0.9999	0.86	2.60	260.02	0.82	1.13	99.89 ± 1.19	1.19	309.64	358.78
	Caffeic acid		10.3	7.81–500	$y = 33.916x + 3.71$	0.9997	0.49	1.48	284.72	1.10	0.93	99.04 ± 0.54	0.55	332.95	384.25
Ethanol	Quinic acid	280	2.3	7.81–500	$y = 257.94x + 14.87$	0.9997	0.32	0.98	127.68	1.04	1.34	99.44 ± 1.08	1.08	176.79	226.33
	Rutin		10.7	7.81–500	$y = 37.918x + 4.54$	0.9993	0.58	1.76	788.64	0.76	0.62	99.54 ± 1.14	1.15	836.94	887.02
	Ferulic acid		24.9	7.81–500	$y = 24.915x + 4.70$	0.9999	0.70	2.13	248.40	1.29	0.56	98.44 ± 0.70	0.71	297.36	348.45
	Epicatechin	320	4.8	7.81–500	$y = 16.919x + 3.46$	0.9998	0.95	2.95	170.46	1.09	1.41	99.22 ± 0.49	0.50	219.52	268.98
	Scopoletin		5.4	7.81–500	$y = 21.003x + 2.60$	0.9999	0.81	2.45	225.89	1.06	1.45	99.55 ± 0.95	0.96	274.47	324.94
	1,4-Dicaffeoylquinic acid		7.2	7.81–500	$y = 33.003x + 2.54$	0.9999	0.51	1.56	446.78	0.84	1.10	99.81 ± 1.04	1.04	495.87	545.57
	Quercetin		13.8	7.81–500	$y = 39.005x + 3.67$	0.9996	0.54	1.74	566.94	0.43	1.33	98.51 ± 0.69	0.70	614.98	666.43
	Apigenin		17.7	7.81–500	$y = 28.998x + 4.58$	0.9999	0.62	1.89	383.49	1.16	0.48	99.37 ± 0.74	0.75	433.14	482.11
Aqueous	Apigenin-7-O-glucuronide	250	14.3	7.81–500	$y = 31.561x + 2.27$	0.9998	0.35	1.05	395.30	0.71	1.08	99.55 ± 1.28	1.29	443.14	495.10
	Hesperidin		17.2	7.81–500	$y = 39.561x + 1.86$	0.9999	0.25	0.76	492.37	0.86	1.32	99.25 ± 1.33	1.34	540.65	591.96
	Malic acid	280	2.7	7.81–500	$y = 23.563x + 5.03$	0.9989	0.94	2.84	104.05	0.95	1.06	98.60 ± 0.69	0.70	152.85	203.78
	p-coumaric acid		14.1	7.81–500	$y = 21.357x + 6.40$	0.9999	1.10	3.34	371.31	1.24	0.73	99.34 ± 0.75	0.75	420.46	471.02
	Luteolin 7-O-glucoside		14.4	7.81–500	$y = 25.582x + 3.01$	0.9999	0.56	1.69	349.81	0.76	0.69	99.44 ± 1.29	1.30	398.28	447.92
	Kaempferol		19.4	7.81–500	$y = 30.954x + 6.51$	0.9999	0.72	2.19	683.09	0.72	1.01	98.88 ± 0.80	0.81	732.51	782.89

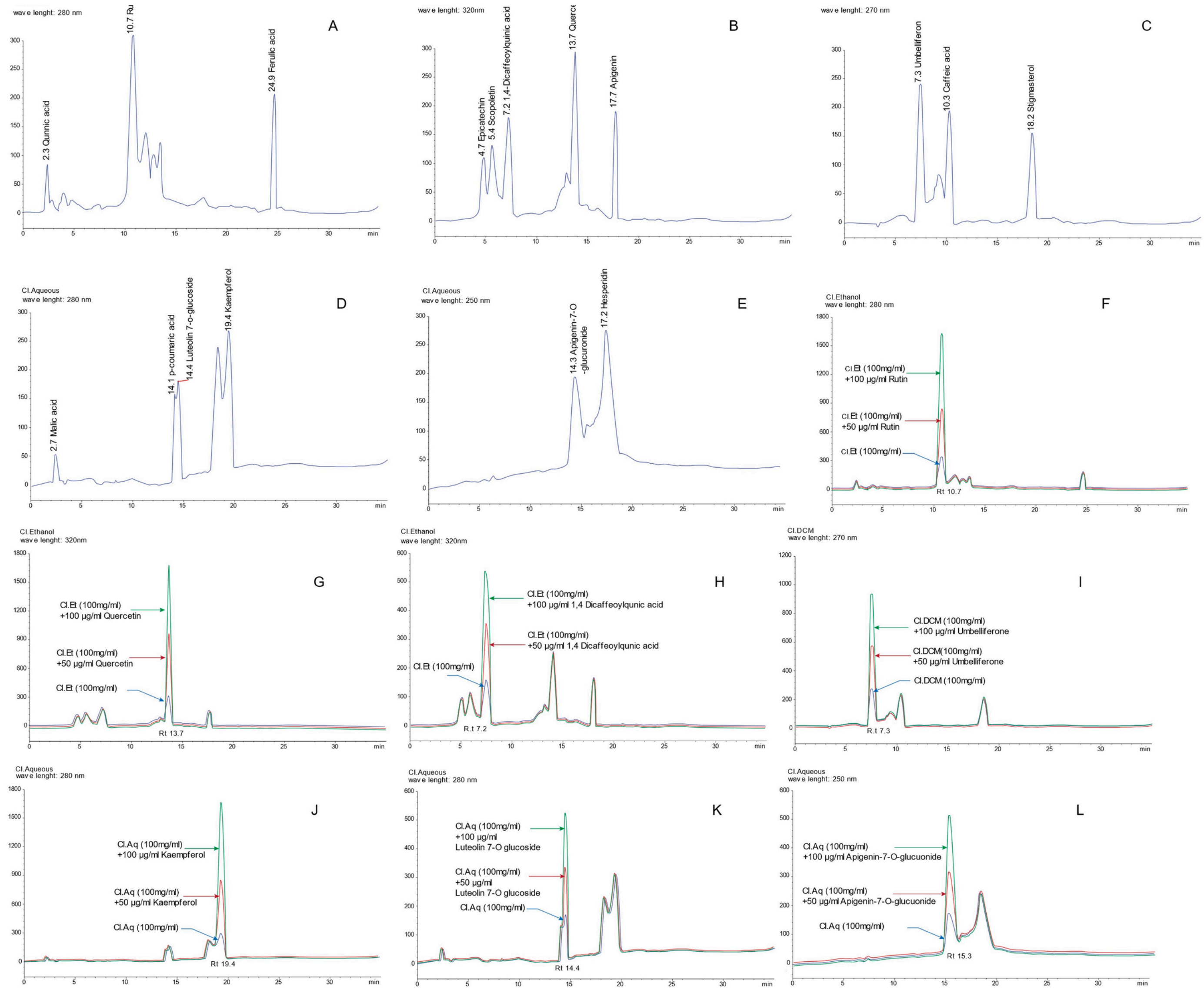


Fig. 2. HPLC DAD-UV/Vis chromatograms of sequential extracts of *C. lanatus* seeds at different wavelengths and standard addition validation chromatograms of bioactive compounds. HPLC chromatograms of ethanol extract (A) quinic acid, rutin, and ferulic acid at wavelength 280 nm, (B) epicatechin, scopoletin, 1,4-dicaffeoylquinic acid, quercetin, and apigenin at wavelength 320 nm; HPLC chromatograms of DCM extract (C) umbelliferone, caffeic acid and stigmasterol at wavelength 270 nm. HPLC chromatograms of aqueous extract (D) malic acid, p-coumaric acid, luteolin 7-O-glucoside, and kaempferol at wavelength 280 nm (E) apigenin-7-O-glucuronide and hesperidin at wavelength 250 nm. The standard addition chromatograms of (F) rutin, (G) quercetin, (H) 1,4-dicaffeoylquinic acid, (I) umbelliferone (J) kaempferol (K) luteolin 7-O-glucoside, and (l) apigenin-7-O-glucuronide.

In silico studies

Network pharmacology analysis

Potential targets Screening: Each compound's possible target profile was obtained from Swiss target prediction and Drugbank databases with a cutoff probability value of more than 0.5. After deleting duplicates, all target genes were collected and kept, including 279 genes for hexane, 181 genes for DCM, 452 genes for ethanol, and 240 genes for aqueous. The databases were searched for target genes associated with gastrointestinal and respiratory diseases using the keywords "diarrhea," "constipation," "irritable bowel syndrome"," asthma", "coughing, bronchitis", and "wheezing". The target genes for gastrointestinal and respiratory illnesses were selected independently, validated, and scored using the VarElect tool. The top 150 genes were saved for further research. The disease target genes and compound targets were intersected into the Venn diagram, yielding 24 intersected possible targets for hexane, 17 for DCM, 25 for ethanol, and 20 for aqueous extracts. These target genes (Tables S3 and S4) were kept for network construction of the KEGG pathway and gene ontology (GO) enrichment.

KEGG and GO analysis: The target genes were subjected to R studio programs for KEGG pathway and gene ontology (GO) enrichment for underlying processes of gastrointestinal and respiratory disorders. Figs. 3 and S2 depict KEGG pathway and gene ontology (GO) enrichment implicated in gastrointestinal and respiratory illnesses (Tables S5 and S6).

Hexane: The target genes of hexane bioactive compounds significantly enriched for both diseases in several GO terms with prominence in G protein-coupled receptor signaling pathway that is modulated by adenylate cyclase, neurotransmitter levels regulation, insulin secretion regulation, smooth muscle contraction, peptide hormone secretion regulation, protein secretion regulation, muscle system process regulation, acetylcholine receptor signaling pathway and, response to acetylcholine.

Hexane bioactive compounds modulate the following KEGG

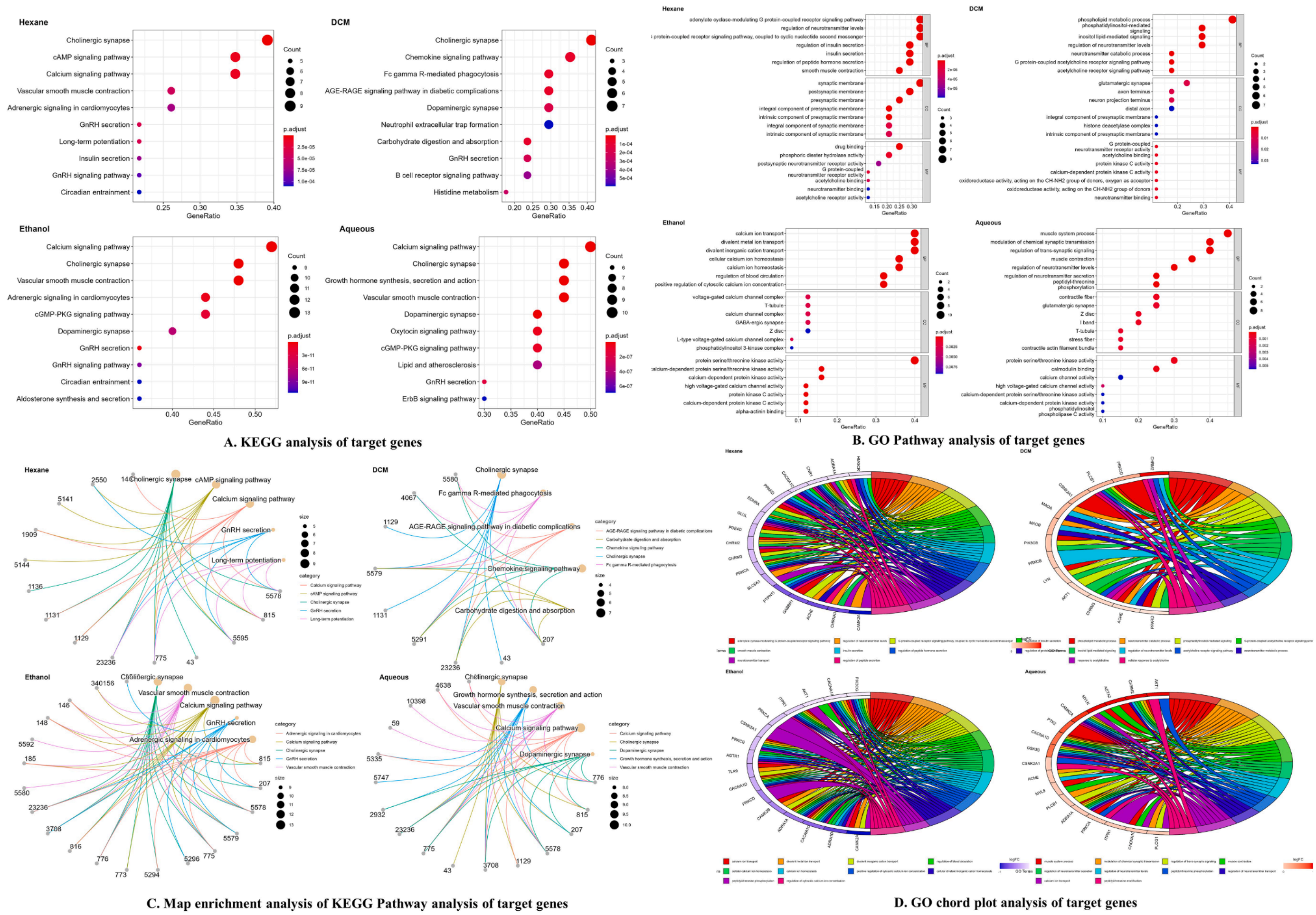


Fig. 3. Bioinformatics analysis of bioactive compounds of sequential extracts of *C. lanatus* seeds for gastrointestinal and respiratory disorders with target pathogenic genes. of Dot plot (**A.**) KEGG pathway and (**B.**) Gene Ontology (GO) enrichment analysis, (**C.**) Map enrichment of KEGG pathway of target genes (**D.**) GO Cochord plot analysis of GO enrichment of target.

signaling pathways; cholinergic synapse, cAMP signaling pathway, calcium signaling pathway, GnRH secretion, Long-term potentiation, vascular smooth muscle contraction, adrenergic signaling in cardiomyocytes, Insulin secretion, GnRH pathway, Parathyroid hormone synthesis, secretion, and action, Dopaminergic synapse, glutamatergic synapse, and neuroactive ligand-receptor interaction.

DCM: For both disorders, the target genes of DCM compounds were prominently shown in various GO pathways, with a focus on phosphatidylinositol mediated signaling, acetylcholine receptor signaling pathway, neurotransmitter levels regulation, neurotransmitter metabolic process, peptidyl-threonine phosphorylation, peptidyl-threonine modification, postsynaptic signal transduction, and regulation of blood vessel diameter.

DCM bioactive compounds modulate the following KEGG signaling pathways: cholinergic synapse, carbohydrate digestion, and absorption, dopaminergic synapse, GnRH secretion, Histidine metabolism, neutrophil extracellular trap formation, B cell receptor signaling pathway, diabetic cardiomyopathy, Insulin resistance, chemokine signaling pathway, serotonergic synapse and, tryptophan metabolism.

Ethanol: The target genes of ethanol compounds were significantly portrayed in numerous GO pathways for both conditions, with a particular emphasis on calcium ion transport, blood circulation regulation, peptidyl-threonine phosphorylation, cytosolic calcium ion concentration regulation, cellular calcium ion homeostasis, cellular response to peptide, platelet activation, second-messenger-mediated signaling and cellular response to peptide hormone stimulus.

Ethanol bioactive compounds modulate the following KEGG signaling pathways; cholinergic synapse, vascular smooth muscle contraction, calcium signaling pathway, oxytocin signaling pathway, GnRH secretion, cardiomyocytes adrenergic signaling, cGMP-PKG signaling pathway, dopaminergic synapse, aldosterone synthesis, and secretion, GnRH signaling pathway, growth hormone synthesis, secretion and action and, Gastric acid secretion.

Aqueous: In several GO pathways, the targeted genes of aqueous bioactive chemicals were significantly enriched for both diseases, with an emphasis on muscle system process, muscle contraction, neurotransmitter secretion regulation, peptidyl-threonine phosphorylation, neurotransmitter transport regulation, calcium ion transport, regulation of muscle system process, blood circulation regulation, neurotransmitter secretion and modulation of chemical synaptic transmission.

Aqueous bioactive compounds modulate the following KEGG signaling pathways were cholinergic synapse, calcium signaling pathway, vascular smooth muscle contraction, oxytocin signaling pathway, GnRH signaling pathway, dopaminergic synapse, cGMP-PKG signaling pathway, Lipid and atherosclerosis, ErbB signaling pathway, cardiomyocytes adrenergic signaling, aldosterone synthesis, and secretion and growth hormone synthesis, secretion and action.

Network Construction: All retained pathogenic targets were subjected to the STRING plugin of Cytoscape for protein-protein interaction (PPI). The gastrointestinal and respiratory pathogenic genes had 53 nodes with 224 edges (Fig. S1). A C-T-D network was constructed in Cytoscape between compounds of bioactive extracts of *C. lanatus* seeds and disease target genes, which had 70 nodes and 117 edges. The network analysis showed that rutin had more potent beneficial effects on targets genes with degree 94, followed by quercetin (degree = 83), kaempferol (degree = 73), ferulic acid (degree = 38), and apigenin-7-glucuronide (degree = 23) (Table S7). Two compound target pathway (C-T-P) network interactions were built for each extract interacting with GO biological processes terms and KEGG signaling pathways (Tables S8, S9, and Fig. 4).

Hexane: GO biological processes terms C-T-P for hexane had 37 nodes and 205 edges in network interaction. The network analysis showed that stearic acid (degree = 48), linoleic acid (degree = 34), oleic acid (degree = 19), and campesterol (degree = 2) had more potent beneficial effects on disease targets genes. The biological process most enriched with target genes and compounds were the G protein-coupled

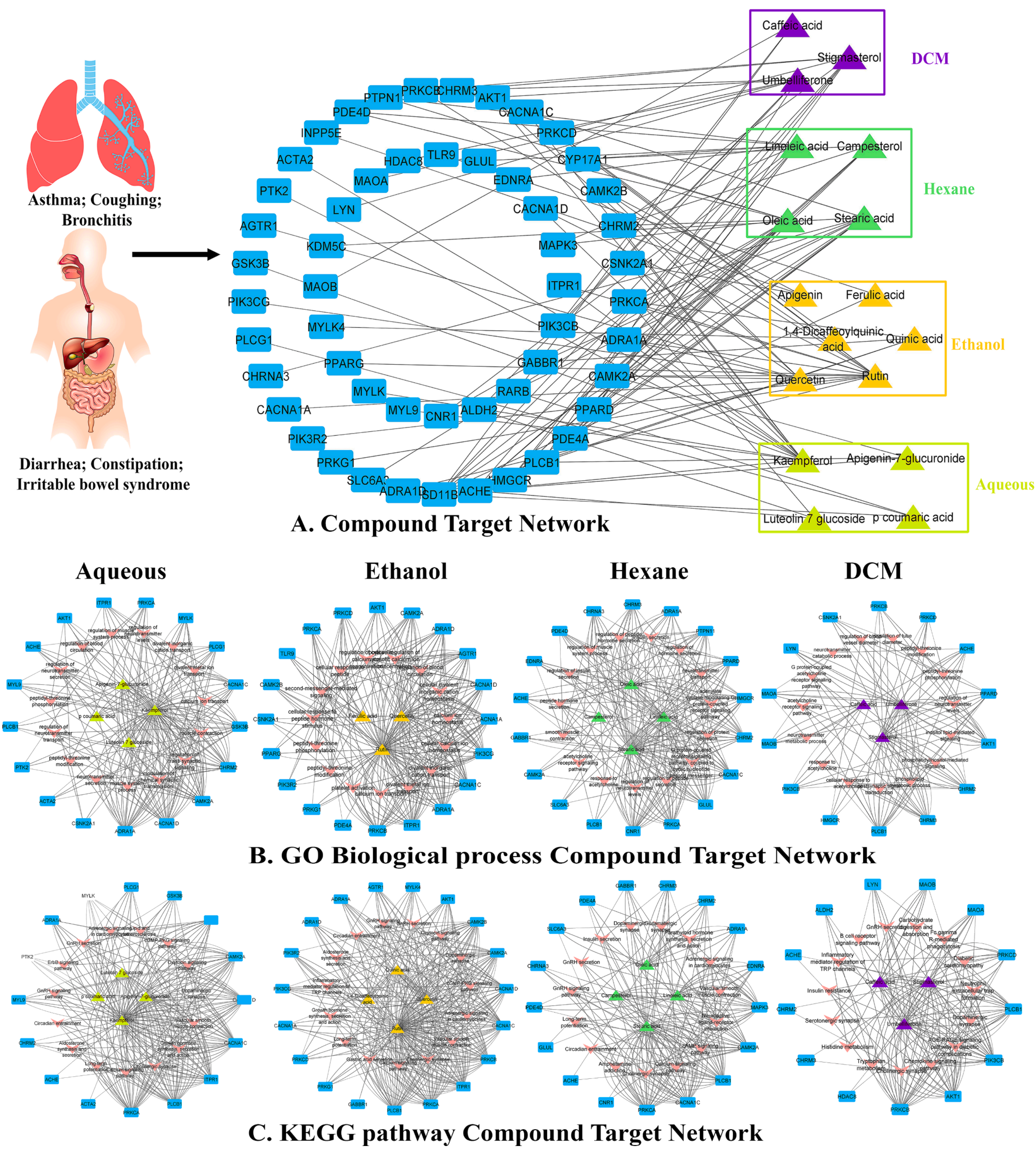


Fig. 4. Network interaction analysis of bioactive compounds. (**A.**) Compound-target network of bioactive compounds and target genes, (**B.**) GO biological process and (**C.**) KEGG pathway network of bioactive compounds and target genes.

receptor signaling pathway (degree = 44), neurotransmitter levels regulation (degree = 28), adenylate cyclase-modulating G protein-coupled receptor signaling pathway (degree = 18), and peptide secretion regulation (degree = 8).

KEGG pathways C-T-P for hexane had 36 nodes and 176 edges. The network analysis showed that stearic acid, linoleic acid, oleic acid, and campesterol had more potent beneficial effects on disease targets genes with degrees 49, 18, 11, and 10, respectively. The KEGG pathway most enriched with target genes and compounds were cholinergic synapse (degree = 39), cAMP signaling pathway (degree = 38), calcium signaling pathway (degree = 18), neuroactive ligand-receptor interaction (degree = 7), and cardiomyocytes adrenergic signaling (degree = 6).

DCM: GO biological processes terms C-T-P for DCM had 32 nodes and 118 edges in network interaction. The network analysis showed that stigmasterol (degree = 53), umbelliferone (degree = 23), and caffeic acid (degree = 6) had more potent beneficial effects on disease targets genes with degrees 33.6 and 16.8. The biological process most enriched with target genes and compounds were regulation of neurotransmitter levels (degree = 35), inositol lipid-mediated signaling (degree = 35), phosphatidylinositol-mediated signaling (degree = 25), and phospholipid metabolic process (degree = 17).

KEGG pathways C-T-P for DCM had 31 nodes and 136 edges in network interaction. The network analysis showed that stigmasterol (degree = 52), umbelliferone (degree = 31), and caffeic acid (degree = 15) had more potent beneficial effects on disease targets genes. The KEGG pathway most enriched with target genes and compounds were cholinergic synapse (degree = 37), serotonergic synapse (degree = 34), chemokine signaling pathway (degree = 6), and diabetic cardiomyopathy (degree = 5).

Ethanol: GO biological processes terms C-T-P for ethanol had 38 nodes and 244 edges in network interaction. The network analysis showed that rutin (degree = 91), quercetin (degree = 77), and ferulic acid (degree = 3) had more potent beneficial effects on disease targets genes. The biological process most enriched with target genes and compounds were cytosolic calcium ion concentration regulation (degree = 68), blood circulation regulation (degree = 62), calcium ion transport (degree = 60), positive regulation of cytosolic calcium ion concentration (degree = 54), and cellular calcium ion homeostasis (degree = 49).

KEGG pathways C-T-P for ethanol had 38 nodes and 290 edges in network interaction. The network analysis showed that rutin (degree = 105), quercetin (degree = 19), quinic acid (degree = 16), and 1,4-dicaffeoylquinic acid (degree = 5) had more potent beneficial effects on disease targets genes. The KEGG pathway most enriched with target genes and compounds were calcium signaling pathway (degree = 73), vascular smooth muscle contraction (degree = 62), cGMP-PKG signaling pathway (degree = 51), cholinergic synapse (degree = 49), and aldosterone synthesis and secretion (degree = 49).

Aqueous: GO biological processes terms C-T-P for aqueous had 36 nodes and 192 edges in network interaction. The network analysis showed that kaempferol (degree = 79), apigenin-7-glucuronide (degree = 13), p coumaric acid (degree = 5), and luteolin 7 glucoside (degree = 4) had more potent beneficial effects on disease targets genes. The biological process most enriched with target genes and compounds were the muscle system process of calcium ion transport (degree = 77), the muscle system process (degree = 69), the regulation of blood circulation (degree = 66), and the regulation of the muscle system process (degree = 63).

KEGG pathways C-T-P for aqueous had 35 nodes and 222 edges in network interaction. The network analysis showed that kaempferol (degree = 78), apigenin-7-glucuronide (degree = 45), luteolin 7 glucoside (degree = 14), and p coumaric acid (degree = 4) had more potent beneficial effects on disease targets genes. The KEGG pathway most enriched with target genes and compounds were the calcium signaling pathway (degree = 70), vascular smooth muscle contraction (degree = 53), cGMP-PKG signaling pathway (degree = 52), and cholinergic synapse (degree = 49).

The network analysis of these interactions defined rutin, apigenin-7-O-glucuronide, quercetin, kaempferol, umbelliferone, luteolin-7-O-glucoside, stearic acid, and linoleic acid had more prominent beneficial activities on target genes with synergistic effects on other genes. These bioactive compounds were studied further for molecular docking to estimate the binding forces and stability of active compounds to target proteins.

Molecular docking

The docking calculations are useful for predicting ligand posture inside the target protein's binding site. The use of physical energy parameters (e.g., solvation energy) in conjunction with a proper force field improves the accuracy of compound docking calculations (Kuhn and Kollman, 2000; Sirous et al., 2019). Bioactive compounds from *C. lanatus* polarity-based sequential extracts were docked with target proteins to see whether they had any positive effects (Fig. 5). The target proteins are members of the calcium-mediated and cholinergic signaling pathways (Figs. S2 and S3) (Sharkey and MacNaughton, 2018). Rutin, quercetin, luteolin-7-O-glucoside, kaempferol, and apigenin-7-O-glucuronide were prominent bioactive compounds with the lowest ligand binding energies for target proteins with the contribution of Vander Waals (G_{vdW}) and lipophilic interactions (G_{Lipo}), as shown in Table 3 and Fig. 6.

Voltage-gated calcium channel β2a: Rutin had dominant binding energies for voltage-gated calcium channels (docking score:−9.167 kcal/mol, $\Delta G_{Binding}$:−50.46 kcal/mol and predicted log Ki:−24.69 μM). It formed hydrogen (HB) interaction with amino acid residues Arg65 (2.18 Å), Arg227 (2.38 Å), Gln380 (2.14 Å), Phe383 (3.05 Å), Val109 (1.68 Å), Val109 (1.73 Å), Ser330 (2.11 Å), Ala89 (2.97 Å), Asp91 (2.00 Å), Asp91 (1.76 Å), Carbon-HB with amino acid Asp384 (2.52 Å), Asp384 (2.74 Å), Tyr402 (2.54 Å), Asp384 (2.87 Å), Val109 (3.07 Å), Tyr402 (2.53 Å), Electrostatic π -Cation interaction with amino acid residue: Arg227 (4.19 Å), Arg227 (3.59 Å), Electrostatic π -Anion Bond Asp384 (4.17 Å), Asp384 (4.63 Å), and π-Alkyl interaction with amino acid residue: Tyr402 (4.10 Å), Tyr406 (5.07). Quercetin (docking score: −5.177 kcal/mol, $\Delta G_{Binding}$:−33.46 kcal/mol and predicted log Ki:−17.3 μM) formed a conventional HB with amino acid Arg227 (2.99 Å), Lys90 (2.95 Å), Asp384 (2.00 Å), Tyr402 (1.73 Å), Val109 (1.67 Å), Electrostatic π -Cation interaction with amino acid residue: Arg227 (3.98 Å), and π- π T-shaped interaction with amino acid residue: Phe92 (5.32). Luteolin 7-glucoside (docking score:−6.938 kcal/mol, $\Delta G_{Binding}$:−36.84 kcal/mol and predicted log Ki: −18.77 μM) formed a conventional HB with amino acid Arg65 (1.99 Å), Phe92h (2.37 Å), Arg227 (2.01 Å), Val109 (1.90 Å), Lys90 (1.94 Å), carbon-HB with amino acid Arg65 (2.51 Å), Val109 (2.60 Å), Val109 (2.39 Å), Electrostatic π -Cation Arg227 (3.87 Å), Electrostatic π -Anion Asp384 (4.12 Å), and π- π T-shaped interaction with amino acid residue: Tyr402 (5.80). Kaempferol (docking score:−5.006 kcal/mol, $\Delta G_{Binding}$:−28.58 kcal/mol and predicted log Ki:−15.18 μM) formed a conventional HB with amino acid Arg227 (2.40 Å), Val109 (1.76 Å), Glu381 (1.73 Å), Electrostatic π -Cation Arg227 (4.24 Å), Electrostatic π -Anion Asp384 (4.06 Å), Asp384 (3.71 Å), and π- π T-shaped interaction with amino acid residue: Tyr402 (5.66). The ranking orders of ligands with VGCC on docking score are given below: rutin, quercetin, luteolin 7-glucoside, kaempferol, apigenin-7-glucuronide, verapamil, stearic acid, and umbelliferone.

Muscarinic M3 receptor (M3, PDB ID: 5ZHP): Rutin had dominant binding energies (docking score:−10.115 kcal/mol, $\Delta G_{Binding}$:−35.3 kcal/mol and predicted log Ki:−18.10 μM) for muscarinic M3 receptor and formed a conventional HB with amino acid Asn103 (2.79 Å), Arg165 (2.75 Å), Arg165 (3.10 Å), Lys484 (3.04 Å), Lys484 (2.02 Å), Lys484 (2.72 Å), Lys487 (2.55 Å), Asn547 (2.42 Å), Thr549 (2.11 Å), Gln97 (2.80 Å), Asn547 (1.80 Å), carbon-HB with amino acid Lys484 (2.44 Å), Thr549 (2.55 Å), Gln97 (2.42 Å), electrostatic attractive charge

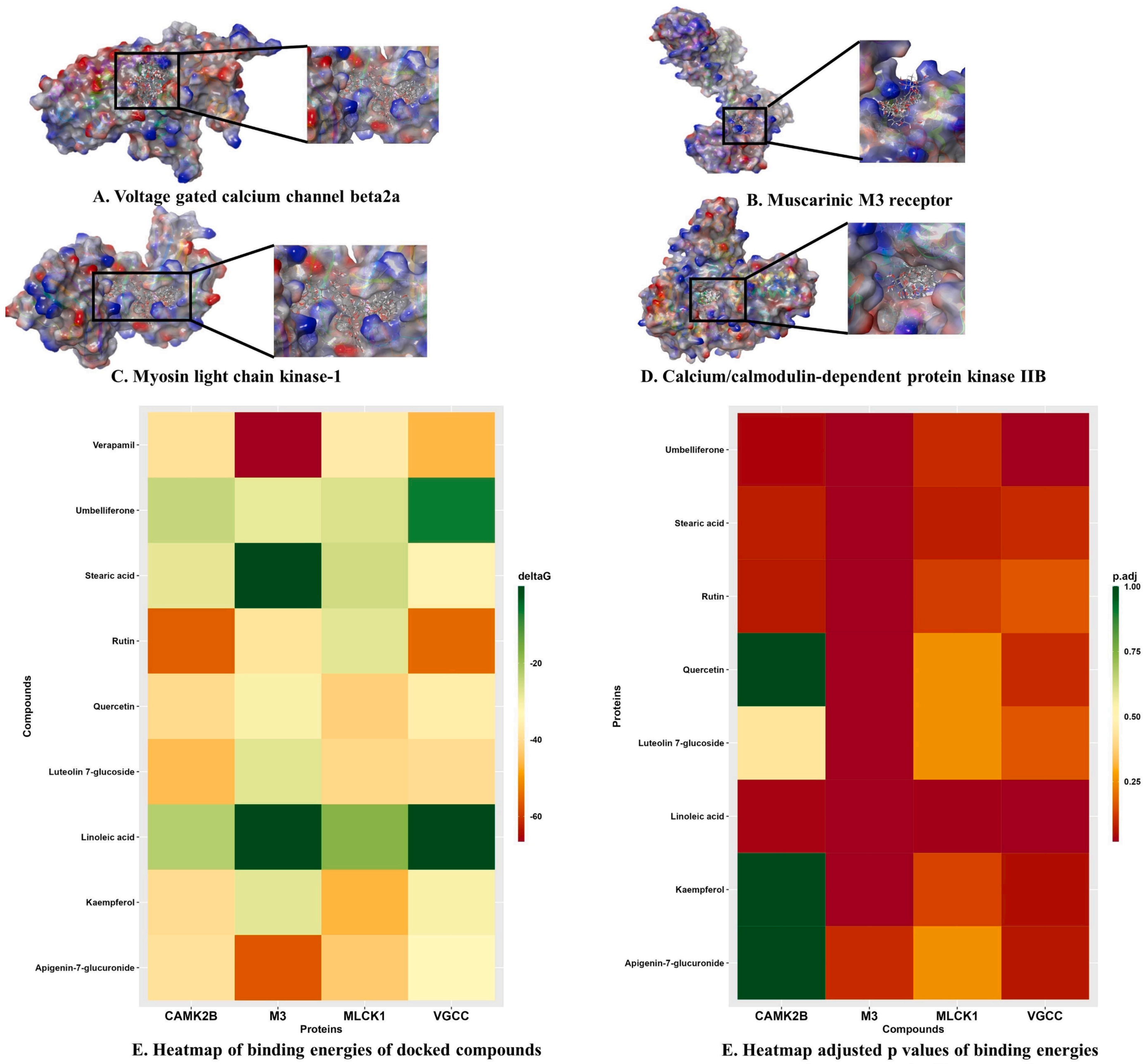


Fig. 5. 3D protein-ligand interaction between bioactive compounds (verapamil, rutin, quercetin, luteolin 7-glucoside, kaempferol, apigenin-7-glucuronide, verapamil, stearic acid, umbelliferone, and linoleic acid) and proteins; **(A.)** voltage-gated calcium channel β2a (VGCC, PDB:1T0J); **(B.)** muscarinic M3 receptor, (M3, PDB:5ZHP) (C) myosin light chain kinase-1 (MLCK-1, PDB:6C6M); and (D) calcium/calmodulin-dependent protein kinase IIB (CAMK2B, PDB: 3BHH). (E) Heatmap of binding energies (kcal/mol) of bioactive compounds and proteins. (F) Heatmap of p-adjusted values of binding energies when compared to verapamil. ($p < 0.05$ vs. verapamil).

interaction with amino acid residue: Arg176 (2.08 Å), Arg165 (5.21 Å), Arg179 (3.01 Å), and π-Alkyl interaction with amino acid residue: Ala488 (4.37). Quercetin (docking score:−6.819 kcal/mol, $\Delta G_{Binding}$:−28.65 kcal/mol and predicted log Ki:−15.21 μM) formed a conventional HB with amino acid Asn103 (2.56 Å), Asn103 (2.20 Å), Arg165 (3.07 Å), Arg165 (2.44 Å), Arg176 (2.26 Å), Ala488 (2.97 Å), π - donor hydrogen-interaction with amino acid residue: Arg165 (3.70 Å), electrostatic π -Cation interaction with amino acid residue: Arg165 (3.70 Å), Arg179 (4.98 Å), Lys484 (4.57 Å), and π-Alkyl interaction with amino acid residue: Ala488 (4.89 Å), Ala488 (4.56 Å), Lys484 (5.45). Kaempferol (docking score:−6.291 kcal/mol, $\Delta G_{Binding}$:−25.97 kcal/mol and predicted log Ki:−14.05 μM) had conventional HB with amino acid Asn103 (2.53 Å), Asn103 (2.34 Å), Arg165 (3.01 Å), Asn547 (2.29 Å), Ala488 (3.04 Å), π - Donor HB with amino acid Arg165 (3.41 Å), electrostatic π -Cation interaction with amino acid residue: Arg165 (3.41 Å), Lys484 (4.58 Å), π-σ interaction with amino acid residue: Lys484 (2.69 Å), and π-Alkyl interaction with amino acid residue: Ala488 (4.65 Å), Ala488 (4.54 Å), Lys484 (4.94 Å), Ala488 (5.43). Apigenin-7-glucuronide (docking score:−5.697 kcal/mol, $\Delta G_{Binding}$:−52.54 kcal/mol and predicted log Ki: −25.59 μM) formed a conventional HB with amino acid Gln97 (2.09 Å), Asn103 (1.91 Å), Arg165 (2.73 Å), Thr549 (2.19 Å), Gln97 (2.73 Å), Lys484 (2.86 Å), carbon-HB with amino acid Thr100 (2.88 Å), Asn103 (2.75 Å), Asn547 (2.57 Å), π - donor hydrogen-interaction with amino acid residue:

Table 3.
Binding energies (kcal/mol) of compounds with Voltage-gated calcium channel beta2a, muscarinic receptor 3 (PDB ID: 5ZHP) Calcium/calmodulin-dependent protein kinase IIB and Myosin light chain kinase-1 calculated by Prime MMGBSA.

Name	Docking score	$\Delta G_{Binding}$	Log K_i (μMolar)	$\Delta G_{Coulomb}$	$\Delta G_{Covalent}$	ΔG_{Hbond}	$\Delta G_{Lipophilic}$	$\Delta G_{Solv\ GB}$	ΔG_{vdW}	Residue-Ligand Interactions with Distance (Å)	
										Hydrogen Bonds	**Electrostatic /Hydrophobic Bonds**
Voltage-gated calcium channel beta2a (PDB:1T0J)											
Rutin	−9.167	−50.46	−24.69	−38.13	7.23	−4.74	−9.73	43.11	−46.31	**Conventional Hydrogen Bond:** Arg65 (2.18), Arg227 (2.38), Gln380 (2.14), Phe383 (3.05), Val109 (1.68), Val109 (1.73), Ser330 (2.11), Ala89 (2.97), Asp91 (2.00), Asp91 (1.76) **Carbon-Hydrogen Bond:** Asp384 (2.52), Asp384 (2.74), Tyr402 (2.54), Asp384 (2.87), Val109 (3.07), Tyr402 (2.53)	**Electrostatic π -Cation Bond:** Arg227 (4.19), Arg227 (3.59), **Electrostatic π -Anion Bond** Asp384 (4.17), Asp384 (4.63), **π-Alkyl Bond**: Tyr402 (4.10), Tyr406 (5.07)
Quercetin	−5.177	−33.46	−17.3	−24.4	0.93	−2.24	−6.75	25.99	−23.47	**Conventional Hydrogen Bond:** Arg227 (2.99), Lys90 (2.95), Asp384 (2.00), Tyr402 (1.73), Val109 (1.67)	**Electrostatic π -Cation Bond:** Arg227 (3.98), **π- π T-shaped Bond:** Phe92 (5.32)
Luteolin 7-glucoside	−6.938	−36.84	−18.77	−27.7	4.94	−3.15	−8.68	36.39	−36.95	**Conventional Hydrogen Bond:** Arg65 (1.99), Phe92h (2.37), Arg227 (2.01), Val109 (1.90), Lys90 (1.94), **Carbon-Hydrogen Bond:** Arg65 (2.51), Val109 (2.60), Val109 (2.39)	**Electrostatic π -Cation** Arg227 (3.87), **Electrostatic π -Anion** Asp384 (4.12), **π- π T-shaped Bond:** Tyr402 (5.80)
Kaempferol	−5.006	−28.58	−15.18	−21.4	1.61	−1.48	−3.77	20.2	−22.62	**Conventional Hydrogen Bond:** Arg227 (2.40), Val109 (1.76), Glu381 (1.73)	**Electrostatic π -Cation** Arg227 (4.24), **Electrostatic π -Anion** Asp384 (4.06), Asp384 (3.71), **π- π T-shaped Bond:** Tyr402 (5.66)
Apigenin-7-glucuronide	−3.636	−30.55	−16.04	−24.42	6.56	−4.14	−6.49	29.5	−31.02	**Conventional Hydrogen Bond:** Gly259 (2.08), Arg424 (2.06), Asp319 (1.83), Asp319 (1.77)	**Electrostatic Attractive Charge:** Lys254 (2.90), Lys254 (2.53), Lys247 (3.44) **π-σ Bond**: Ile263 (2.91), **π-Alkyl Bond**: Ile263 (4.18), Ile261 (5.13)
Verapamil	−2.351	−42.52	−15.24	−16.52	1.27	−1.14	−14.39	33.32	−44.34	**Conventional Hydrogen Bond:** Arg227 (2.65) **Carbon Hydrogen Bond:** Tyr402 (2.72), Asp384 (2.54), Tyr402 (2.69), Glu111 (2.61), Ser330 (2.75), Pro336 (2.49), Glu381	**Electrostatic Attractive Charge:** Asp384 (4.51) **Electrostatic π-Cation:** Arg227 (4.00) **Electrostatic π-Anion:** Asp384 (3.66) **Alkyl:** Ala335 (4.32), Ala405

(*continued on next page*)

Table 3. (*continued*)

										(2.67), Ser382 (2.43)	(3.99), Lys110 (4.24), Pro326 (5.46), Ile338 (4.46) **π-Alkyl:** Phe92 (5.37), Lys110 (5.11), Ala409 (4.44)
Stearic acid	−1.191	−32.5	−16.89	−6.55	1.83	−0.45	−10.97	11.22	−27.58	**Conventional Hydrogen Bond:** Gly259 (2.06)	**Alkyl Bond**: Ile263 (5.06), Lys254 (4.57), Ile261 (4.54), Ile263 (4.34), Ile263 (5.29), Arg265 (5.13), Arg265 (3.94), Arg265 (4.18)
Umbelliferone	−0.498	−6.43	−5.56	−28.02	0.27	−1.07	−3.97	46.57	−19.27	**Conventional Hydrogen Bond:** Gly259 (1.93), **π - Donor H Bond:** Ile263 (3.31)	**π-Alkyl Bond**: Ile261 (5.43), Ile263 (4.55), Ile261 (4.96)
Muscarinic M3 receptor (M3, PDB ID: 5ZHP)											
Rutin	−10.115	−35.3	−18.10	−283.61	10.59	−5.79	−10.21	297.94	−40.5	**Conventional Hydrogen Bond:** Asn103 (2.79), Arg165 (2.75), Arg165 (3.10), Lys484 (3.04), Lys484 (2.02), Lys484 (2.72), Lys487 (2.55), Asn547 (2.42), Thr549 (2.11), Gln97 (2.80), Asn547 (1.80), **Carbon-Hydrogen Bond:** Lys484 (2.44), Thr549 (2.55), Gln97 (2.42)	**Electrostatic Attractive Charge:** Arg176 (2.08), Arg165 (5.21), Arg179 (3.01), **π-Alkyl Bond:** Ala488 (4.37)
Quercetin	−6.819	−28.65	−15.21	−11.56	2.26	−3.1	−1.85	21.89	−29.99	**Conventional Hydrogen Bond:** Asn103 (2.56), Asn103 (2.20), Arg165 (3.07), Arg165 (2.44), Arg176 (2.26), Ala488 (2.97), **π - Donor H Bond:** Arg165 (3.70)	**Electrostatic π -Cation:** Arg165 (3.70), Arg179 (4.98), Lys484 (4.57), **π-Alkyl Bond:** Ala488 (4.89), Ala488 (4.56), Lys484 (5.45)
Kaempferol	−6.291	−25.97	−14.05	−12.91	2.24	−1.71	−1.81	23.61	−29.48	**Conventional Hydrogen Bond:** Asn103 (2.53), Asn103 (2.34), Arg165 (3.01), Asn547 (2.29), Ala488 (3.04) **π - Donor H Bond:** Arg165 (3.41),	**Electrostatic π -Cation:** Arg165 (3.41), Lys484 (4.58) **π-σ Bond**: Lys484 (2.69), **π-Alkyl Bond:** Ala488 (4.65), Ala488 (4.54), Lys484 (4.94), Ala488 (5.43)
Apigenin-7-glucuronide	−5.697	−52.54	−25.59	−190.98	6.6	−4.72	−8.99	194.26	−43.92	**Conventional Hydrogen Bond:** Gln97 (2.09), Asn103 (1.91), Arg165 (2.73), Thr549 (2.19), Gln97 (2.73), Lys484 (2.86), **Carbon-Hydrogen Bond:** Thr100 (2.88), Asn103 (2.75), Asn547 (2.57) **π - Donor H Bond:** Val101 (3.13)	**Electrostatic Attractive Charge:** Lys487 (2.51), Lys484 (5.59 **Electrostatic π -Cation Bond:** Arg165 (3.87), Arg179 (4.39) **π-Alkyl Bond:** Ala488 (5.31), Val101 (4.64), Arg179 (5.44)
Umbelliferone	−4.43	−26.55	−14.30	−4.56	0.37	−1.08	−7.54	9.51	−18.55		**π-Alkyl Bond:** Ala488 (5.02),

(*continued on next page*)

Table 3. (*continued*)

										Conventional Hydrogen Bond: Asn103 (1.86)	Leu106 (5.45), Ala488 (4.41)
Luteolin 7-glucoside	−3.103	−25.67	−13.92	−23.12	10.96	−4.15	−4.06	31.44	−33.3	**Conventional Hydrogen Bond:** Asn103 (2.57), Asn103 (1.80), Arg165 (1.80), Arg176 (2.07), Lys487 (2.75), Asn547 (2.73), Thr100 (2.63)	Lys487 (4.11)
Verapamil	−2.264	−61.39	−23.43	2.45	4.92	−1.22	−27.11	2.8	−41.13	**Conventional Hydrogen Bond:** Ile222 (2.45), Lys522 (2.03), **Carbon Hydrogen Bond:** Phe221 (2.73), Lys522 (2.42), Asn513 (3.04), Glu219 (2.69), Glu219 (2.63), Cys220 (3.03), Cys220 (2.66), Ile222 (2.53)	**Electrostatic π-Cation:** Trp525 (4.66), Trp525 (4.11),
Calcium/calmodulin-dependent protein kinase IIB (CAMK2B, PDB: 3BHH)											
Rutin	−10.314	−51.84	−25.28	−72.4	3.82	−4.78	−7.71	74.39	−44.9	**Conventional Hydrogen Bond:** Asn256 (2.43), Gly179 (1.97), Trp215 (2.05), Glu217 (1.68), Glu217 (2.46), **Carbon-Hydrogen Bond:** Gly179 (2.49)	**Electrostatic π -Cation Bond:** Arg298 (3.51), Arg298 (3.35), **π-Alkyl Bond**: Pro212 (5.17)
Luteolin 7-glucoside	−7.605	−41.9	−20.97	−28.51	13.55	−5.46	−14.92	40.57	−39.7	**Conventional Hydrogen Bond:** Arg66 (2.94), Arg66 (1.90), Lys293 (1.97), Lys293 (2.22), Pro212 (1.73), Leu67 (2.27), Arg298 (2.96), Arg297 (2.63) **π - Donor H Bond:** Arg66 (3.80), Arg66 (2.99)	**Electrostatic π -Cation Bond:** Arg66 (3.80), Arg66 (3.00) **π- π T-shaped Bond:** Trp215 (4.94), Trp215 (4.96), **Amide- π Stacked:** Trp215 (4.96), Phe214 (4.96), **π-Alkyl Bond**: Pro212 (4.13), Pro212 (4.38), Arg297 (4.71), Lys69 (4.39), Pro212 (5.36)
Apigenin-7-glucuronide	−6.68	−35.67	−18.26	−19.19	14.42	−5.06	−14.4	36.46	−40.61	**Conventional Hydrogen Bond:** Arg66 (2.11), Lys293 (1.92), Lys293 (2.15), Arg298 (3.08), Leu67 (2.76), Arg297 (2.45), Arg298 (3.00) **π - Donor H Bond:** Arg66 (2.66), Arg66 (2.92)	**Electrostatic Attractive Charge:** Arg66 (2.66), Arg66 (1.83), Arg297 (4.77) **Electrostatic π -Cation Bond:** Arg66 (3.79), Arg66 (3.00), **π- π Stacked Bond:** Phe294 (5.17), **π- π T-shaped Bond:** Trp215 (4.92), Trp215 (4.96) **Amide- π Stacked:** Phe214 (4.83), Trp215 (4.83) **π-Alkyl Bond**: Pro212 (4.19), Pro212 (4.37), Arg297 (4.73), Lys69 (4.47), Pro212 (5.29)

(*continued on next page*)

Table 3. (*continued*)

Quercetin	−6.096	−37.09	−18.88	−33.16	2.65	−2.76	−5.92	33.67	−28.07	**Conventional Hydrogen Bond:** Asn256 (2.94), Lys138 (2.66), Trp215 (2.32), Arg298 (2.18), Arg298 (2.46), Arg298 (1.89), Glu140 (2.54), Thr177 (3.01), Arg297 (1.95), Glu189 (2.04), Glu189 (1.93) **Carbon-Hydrogen Bond:** Pro257 (2.97), Pro178 (2.59), Arg297 (2.59), Arg298 (2.66), Arg297 (2.70)	**Electrostatic Attractive Charge:** Arg66 (5.59) **π -Alkyl Bond**: Trp215 (5.16), Pro178 (4.80)
Kaempferol	−5.863	−36.82	−18.76	−22.12	4.48	−1.9	−10.67	25.75	−26.3	Trp215 (1.86), Glu217 (1.74), Phe294 (2.13) **π - Donor H Bond:** Trp215 (2.44),	**Electrostatic π -Cation Bond:** Arg66 (3.87), Arg298 (4.37), **π- π T-shaped Bond:** Trp215 (4.32), **π-Alkyl Bond**: Arg297 (4.94), Arg298 (4.83)
Umbelliferone	−5.033	−22.7	−12.63	−16.01	0.81	−1.09	−7.06	26.47	−25.31	**Conventional Hydrogen Bond:** Asp232 (2.02), **Carbon-Hydrogen Bond:** Lys69 (2.81), Tyr231 (2.96	**π-σ Bond**: Lys69 (2.59), **π-Alkyl Bond**: Lys69 (4.68)
Stearic acid	−2.878	−26.19	−14.15	−16.93	2.88	−1.49	−13.69	37.01	−33.97	**Conventional Hydrogen Bond:** Arg187 (2.37), Lys227 (1.65	**Alkyl Bond**: Arg187 (5.11), Ile255 (5.18), Ala258 (3.76), Ile255 (5.01
Linoleic acid	−2.349	−20.8	−11.80	26.68	11.8	−2.35	−22.39	0.16	−34.7	**Conventional Hydrogen Bond:** Arg66 (1.84), Arg66 (2.64), Trp215 (2.20)	**Electrostatic Attractive Charge:** Arg66 (4.10), Arg298 (5.37 **Alkyl Bond**: Arg66 (4.33), Lys69 (4.52), Pro212 (4.88), Pro212 (4.91), Pro212 (5.31), Arg297 (4.67), Arg297 (4.86), Lys69 (3.99), Pro234 (5.03) **π-Alkyl Bond**: Trp215 (4.55), Trp215 (5.28), Trp215 (5.50), Trp215 (5.38), Phe294 (4.98)
Verapamil	−1.537	−36.04	−12.42	26.1	−0.27	−1.06	−15.28	1.48	−44.54	**Carbon Hydrogen Bond:** Arg298 (2.82), Glu217 (2.48), Asp216 (2.53), Glu82 (2.98), Glu82 (2.99) **π -Donor Hydrogen Bond:** Asn256 (2.93)	**Electrostatic π-Cation:** Arg298 (3.44) **π - π T-Shaped:** Trp215 (4.88) **Alkyl:** Ala258 (4.48), Pro212 (3.94) **π -Alkyl:** Ala258 (5.05)
Myosin light chain kinase-1 (MLCK-1, PDB: 6C6M)											
Quercetin	−6.991	−39.01	−19.71	−45.28	11.4	−3.27	−8.72	34.3	−26.1	**Conventional Hydrogen Bond:** Arg456 (2.38), Arg456 (2.28), Glu450 (1.70), Arg480 (2.59),	**Electrostatic π -Cation Bond:** Arg480 (3.96), **π-Alkyl Bond**: Arg480 (5.23), Arg480 (5.02)

(*continued on next page*)

Table 3. (*continued*)

										Arg4802 (2.76), Leu449 (2.19), Asp481 (1.83), Asp481 (2.51), Glu458 (2.20), Glu450 (1.92), Ser482 (3.05), **Carbon-Hydrogen Bond:** Val454 (2.73), Asp481 (2.78), Arg480 (2.49)	
Rutin	−6.94	−25.86	−14	−134.92	15.06	−6.01	−10.16	142.37	−29.89	**Conventional Hydrogen Bond:** Glu417 (1.83), His470 (1.97), Val454 (1.67), Trp447 (1.84), **Carbon-Hydrogen Bond:** Trp447 (2.95)	**Electrostatic π -Cation Bond:** Arg456 (4.32), **Electrostatic π -Anion Bond:** Glu444 (4.79), **π-Alkyl Bond**: Val463 (5.20), Val463 (5.40), Ala446 (4.15)
Kaempferol	−5.29	−42.83	−21.37	−34.11	2.02	−2.15	−9.6	30.56	−28.39	**Conventional Hydrogen Bond:** Arg456 (2.18), His470 (1.85), Trp447 (1.66), **π - Donor H Bond:** Trp447 (2.92)	**π-Alkyl Bond**: Val463 (4.56), Val463 (5.03), Ala446 (4.15), Val463 (5.44)
Apigenin-7-glucuronide	−5.023	−39.88	−20.09	−46.91	6.6	−4.77	−5.65	37.09	−26.16	**Conventional Hydrogen Bond:** Gln421 (2.06), Arg434 (2.38), Arg434 (2.20), Arg434 (2.55), Arg434 (3.05), Ser499 (1.99), Lys432 (1.76), Lys419 (2.76), Thr430 (1.77), **Carbon-Hydrogen Bond:** Pro420 (2.56), Phe433 (2.42), Arg434 (2.57)	**Electrostatic Attractive Charge:** Arg434 (1.83) **Electrostatic π -Cation Bond:** Lys432 (4.37), Lys432 (4.71) **π-Alkyl Bond**: Lys432 (4.70), Lys432 (4.92), Lys432 (5.27
Linoleic acid	−3.745	−16.18	−9.80	−6.93	1.91	−2.34	−6.77	24.4	−26.45	**Conventional Hydrogen Bond:** Arg456 (2.07), Arg456 (1.86)	**Electrostatic Attractive Charge:** Arg456 (3.96) **Alkyl Bond**: Leu449 (4.33), Ile461 (5.31)
Stearic acid	−3.322	−23.45	−12.96	−10.13	5.3	−2.82	−12.49	25.05	−28.35	**Conventional Hydrogen Bond:** Arg456 (1.91)	**Electrostatic Attractive Charge:** Arg456 (1.85) **Alkyl Bond**: Leu449 (4.58), Ile461 (5.28)
Umbelliferone	−3.048	−24.74	−13.52	−10.08	0.54	−0.71	−7.68	17.21	−23.96	**Conventional Hydrogen Bond:** Val454 (1.84), Trp447 (2.64), **π - Donor H Bond:** Trp447 (3.16)	**π-Alkyl Bond**: Ala446 (5.24), Val463 (4.62), Ala446 (4.27), Val463 (5.22)
Luteolin 7-glucoside	−2.204	−37.67	−19.13	−34.18	7.31	−4.26	−9.29	44.64	−40.16	**Conventional Hydrogen Bond:** Val445 (2.78), Arg456 (2.22), Glu465 (2.14), Glu465 (2.27), Glu417 (2.02), **Carbon-Hydrogen Bond:** Glu444 (2.78), Tyr464 (2.37)	**π-Alkyl Bond**: Ala446 (4.76), Val463 (5.01)
Verapamil	−2.844	−33.73	−11.42	7.71	7.06	−1.13	−14.36	4.17	−36.53	**Conventional Hydrogen Bond:** Arg456 (2.49), Arg456 (2.03) **Carbon Hydrogen**	**Electrostatic π-Cation:** Arg480 (4.17) **Alkyl:** Leu449

(*continued on next page*)

Table 3. (*continued*)

	Bond: Leu449 (2.93), Glu450 (2.89), Glu465 (2.55), Glu465 (2.59), Asp481 (2.54), Pro453 (2.63) **π -Donor Hydrogen Bond:** Arg480 (4.17)	(4.89), Val454 (4.56), Ile461 (5.25)

$\Delta G_{Binding}$: Binding free energy, **Log Ki:** Logarithmic of Inhibition Constant (K_i), **$\Delta G_{Coulomb}$:** Coulomb binding energy, **$\Delta G_{Covalent}$:** Covalent binding energy **ΔG_{Hbond}:** Hydrogen bonding energy, **$\Delta G_{Lipophilic}$:** Lipophillic binding energy, **$\Delta G_{Solv\ GB}$:** Generalized born electrostatic solvation energy **ΔG_{vdW}:** Van der Waals forces energy. These all energy contribute to Binding free energy (**$\Delta G_{Binding}$**).

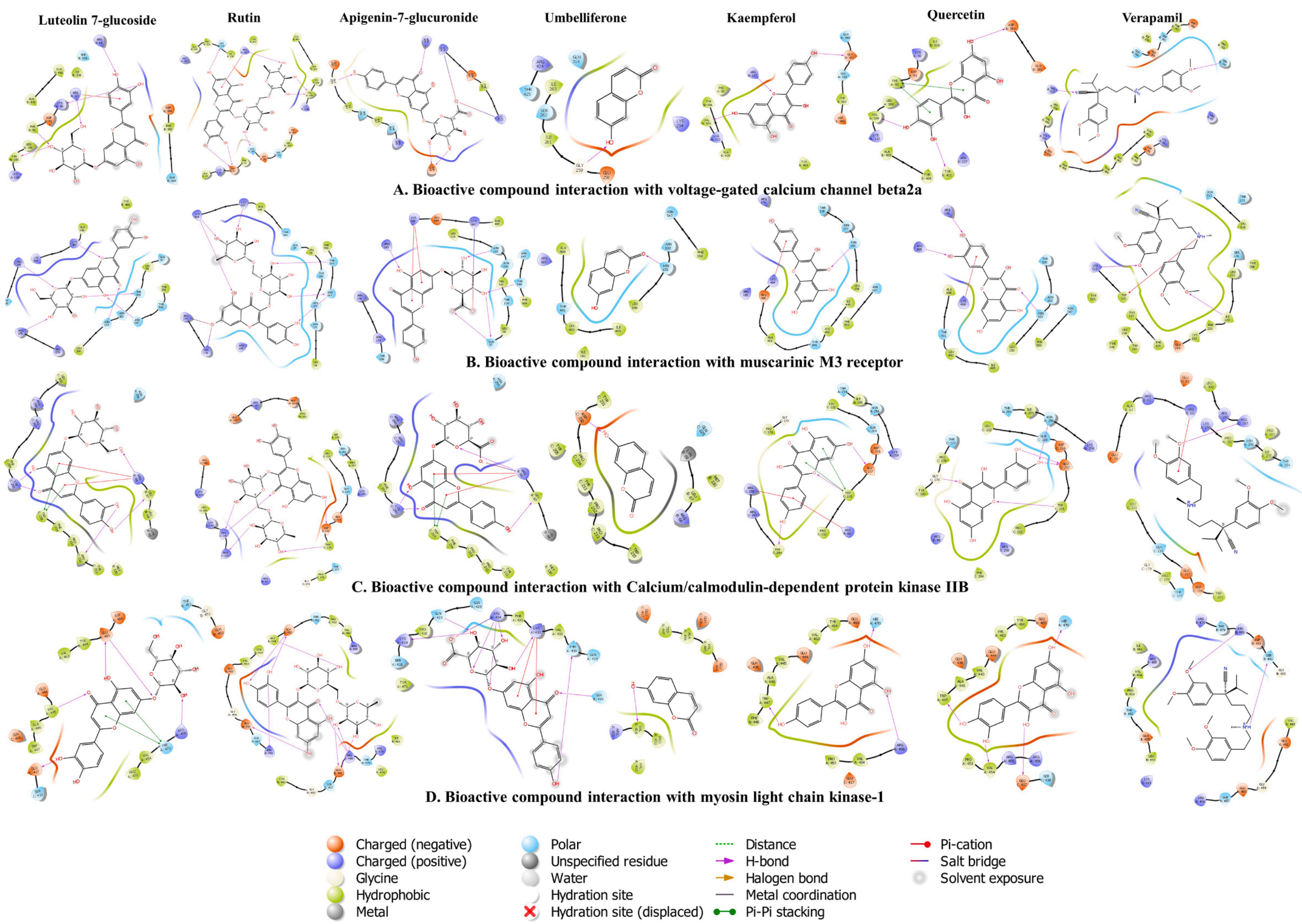


Fig. 6. 2D protein-ligand interaction between bioactive compounds (verapamil, rutin, quercetin, luteolin 7-glucoside, kaempferol, apigenin-7-glucuronide, verapamil, stearic acid, umbelliferone, and linoleic acid) and proteins; **(A.)** voltage-gated calcium channel β2a (VGCC, PDB:1T0J); **(B.)** muscarinic M3 receptor, (M3, PDB:5ZHP) **(C.)** myosin light chain kinase-1 (MLCK-1, PDB:6C6M); and **(D.)** calcium/calmodulin-dependent protein kinase IIB (CAMK2B, PDB: 3BHH).

Val101 (3.13), Electrostatic Attractive Charge interaction with amino acid residue: Lys487 (2.51 Å), Lys484 (5.59, Electrostatic π -Cation interaction with amino acid residue: Arg165 (3.87 Å), Arg179 (4.39 Å), and π-Alkyl interaction with amino acid residue: Ala488 (5.31 Å), Val101 (4.64 Å), Arg179 (5.44). The ranking orders of ligands with CAMK2B based on docking score are below; rutin, quercetin, kaempferol, apigenin-7-glucuronide, umbelliferone, luteolin 7-glucoside, and verapamil.

Calcium/calmodulin-dependent protein kinase IIB: Rutin scored binding affinity (docking score:−10.314 kcal/mol, $\Delta G_{Binding}$:−51.84 kcal/mol and predicted log Ki:−25.28 μM) for calcium/calmodulin-dependent protein kinase IIB (−39.01 kcal/mol). It formed a Conventional HB with amino acid Asn256 (2.43 Å), Gly179 (1.97 Å), Trp215 (2.05 Å), Glu217 (1.68 Å), Glu217 (2.46 Å), Carbon-HB with amino acid Gly179 (2.49 Å), Electrostatic π -Cation interaction with amino acid residue: Arg298 (3.51 Å), Arg298 (3.35 Å), and π-Alkyl interaction with amino acid residue: Pro212 (5.17). Luteolin 7-glucoside (docking score:−7.605 kcal/mol, $\Delta G_{Binding}$:−41.9 kcal/mol and predicted log Ki:−20.97 μM) formed Conventional HB with amino acid Arg66 (2.94 Å), Arg66 (1.90 Å), Lys293 (1.97 Å), Lys293 (2.22 Å), Pro212 (1.73 Å), Leu67 (2.27 Å), Arg298 (2.96 Å), Arg297 (2.63 Å), π - donor HB with amino acid Arg66 (3.80 Å), Arg66 (2.99 Å), Electrostatic π -Cation interaction with amino acid residue: Arg66 (3.80 Å), Arg66 (3.00 Å), π- π T-shaped interaction with amino acid residue: Trp215 (4.94 Å), Trp215

(4.96 Å), Amide- π Stacked interaction with amino acid residue: Trp215 (4.96 Å), Phe214 (4.96 Å), and π-Alkyl interaction with amino acid residue: Pro212 (4.13 Å), Pro212 (4.38 Å), Arg297 (4.71 Å), Lys69 (4.39 Å), Pro212 (5.36). Apigenin-7-glucuronide (docking score:−6.68 kcal/mol, $\Delta G_{Binding}$:−35.67 kcal/mol and predicted log Ki:−18.26 μM) formed Conventional HB with amino acid Arg66 (2.11 Å), Lys293 (1.92 Å), Lys293 (2.15 Å), Arg298 (3.08 Å), Leu67 (2.76 Å), Arg297 (2.45 Å), Arg298 (3.00 Å), π - donor HB with amino acid Arg66 (2.66 Å), Arg66 (2.92 Å), Electrostatic Attractive Charge interaction with amino acid residue: Arg66 (2.66 Å), Arg66 (1.83 Å), Arg297 (4.77 Å), Electrostatic π -Cation interaction with amino acid residue: Arg66 (3.79 Å), Arg66 (3.00 Å), π- π Stacked interaction with amino acid residue: Phe294 (5.17 Å), π- π T-shaped interaction with amino acid residue: Trp215 (4.92 Å), Trp215 (4.96 Å), Amide- π Stacked interaction with amino acid residue: Phe214 (4.83 Å), Trp215 (4.83 Å), and π-Alkyl interaction with amino acid residue: Pro212 (4.19 Å), Pro212 (4.37 Å), Arg297 (4.73 Å), Lys69 (4.47 Å), Pro212 (5.29). Quercetin (docking score:−6.096 kcal/mol, $\Delta G_{Binding}$:−37.09 kcal/mol and predicted log Ki:−18.88 μM) formed Conventional HB with amino acid Asn256 (2.94 Å), Lys138 (2.66 Å), Trp215 (2.32 Å), Arg298 (2.18 Å), Arg298 (2.46 Å), Arg298 (1.89 Å), Glu140 (2.54 Å), Thr177 (3.01 Å), Arg297 (1.95 Å), Glu189 (2.04 Å), Glu189 (1.93 Å), Carbon-HB with amino acid Pro257 (2.97 Å), Pro178 (2.59 Å), Arg297 (2.59 Å), Arg298 (2.66 Å), Arg297 (2.70 Å), Electrostatic Attractive Charge interaction with amino acid residue: Arg66 (5.59 Å), and π -Alkyl Bond: Trp215 (5.16 Å), Pro178 (4.80). The ranking orders of ligands with CAMK2B based on docking score are below; rutin, luteolin 7-glucoside, apigenin-7-glucuronide, quercetin, kaempferol, umbelliferone, stearic acid, linoleic acid, and verapamil.

Myosin light chain kinase-1: Quercetin was predicted with binding energy (docking score:−6.991 kcal/mol, $\Delta G_{Binding}$:−39.01 kcal/mol and predicted log Ki:−19.71 μM) and formed Conventional HB with amino acid Arg456 (2.38 Å), Arg456 (2.28 Å), Glu450 (1.70 Å), Arg480 (2.59 Å), Arg4802 (2.76 Å), Leu449 (2.19 Å), Asp481 (1.83 Å), Asp481 (2.51 Å), Glu458 (2.20 Å), Glu450 (1.92 Å), Ser482 (3.05 Å), Carbon-HB with amino acid Val454 (2.73 Å), Asp481 (2.78 Å), Arg480 (2.49) Electrostatic π -Cation interaction with amino acid residue: Arg480 (3.96 Å), and π-Alkyl interaction with amino acid residue: Arg480 (5.23 Å), Arg480 (5.02). Rutin (docking score:−6.94 kcal/mol, $\Delta G_{Binding}$:−25.86 kcal/mol and predicted log Ki:−14 μM) formed Conventional HB with amino acid Glu417 (1.83 Å), His470 (1.97 Å), Val454 (1.67 Å), Trp447 (1.84 Å), Carbon-HB with amino acid Trp447 (2.95 Å), Electrostatic π -Cation interaction with amino acid residue: Arg456 (4.32 Å), Electrostatic π -Anion interaction with amino acid residue: Glu444 (4.79 Å), and π-Alkyl interaction with amino acid residue: Val463 (5.20 Å), Val463 (5.40 Å), Ala446 (4.15). Kaempferol (docking score:−5.29 kcal/mol, $\Delta G_{Binding}$:−42.83 kcal/mol and predicted log Ki:−21.37 μM) formed Conventional HB with amino acid Arg456 (2.18 Å), His470 (1.85 Å), Trp447 (1.66 Å), π - donor HB with amino acid Trp447 (2.92) and π-Alkyl interaction with amino acid residue: Val463 (4.56 Å), Val463 (5.03 Å), Ala446 (4.15 Å), Val463 (5.44 Å). Apigenin-7-glucuronide (docking score:−5.023 kcal/mol, $\Delta G_{Binding}$: −39.88 kcal/mol and predicted log Ki:−20.09 μM) formed Conventional HB with amino acid Gln421 (2.06 Å), Arg434 (2.38 Å), Arg434 (2.20 Å), Arg434 (2.55 Å), Arg434 (3.05 Å), Ser499 (1.99 Å), Lys432 (1.76 Å), Lys419 (2.76 Å), Thr430 (1.77 Å), carbon-HB with amino acid Pro420 (2.56 Å), Phe433 (2.42 Å), Arg434 (2.57 Å), electrostatic attractive charge interaction with amino acid residue: Arg434 (1.83 Å), Electrostatic π -Cation interaction with amino acid residue: Lys432 (4.37 Å), Lys432 (4.71 Å), and π-Alkyl interaction with amino acid residue: Lys432 (4.70 Å), Lys432 (4.92 Å), Lys432 (5.27). The ranking orders of ligands with MLCK-1 based on docking score are given below, Quercetin, Rutin, Kaempferol, apigenin-7-glucuronide, linoleic acid, stearic acid, umbelliferone, luteolin 7-glucoside, and verapamil.

In vitro experiments

Effects on isolated rabbit jejunum preparation

The antispasmodic effect of polarity-based sequential extracts of *C. lanatus* seeds was investigated using spontaneously contracting jejunum preparations (Fig. 7). The polarity-based sequential extracts exerted concentration-dependent spasmogenic and spasmolytic effects and exhibited concentration-dependent partial or complete inhibitory responses for evoked contractions for K^+ (80 mM) and K^+ (25 mM). When Cl.Hexane and Cl.Aq extracts were subjected to rhythmic contraction of jejunum preparations, spontaneous contraction was suppressed in a concentration-dependent manner with respective EC_{50} 0.3689 mg/ml (95%CI: 0.2258–0.6418 mg/ml) and 1.018 mg/ml (95% CI: 0.7131–1.494 mg/ml). Whereas Cl.DCM and Cl.Et extracts displayed concentration-dependent contractile response (Fig. 7A). Atropine (1 μM) pretreated jejunum preparation diminished the spasmogenic response of Cl.DCM and Cl.Et extracts and showed the concentration-dependent spasmolytic effect with respective EC_{50} 0.3418 mg/ml (95%CI: 0.2121–0.5797 mg/ml) and 0.7091 mg/ml (95%CI: 0.4979–1.028 mg/ml).

In jejunum preparation, the polarity-based sequential extracts of *C. lanatus* seeds were tested on persistent contractions of K^+ (80 mM) and K^+ (25 mM). The polarity-based sequential extracts of *C. lanatus* seeds exhibited the concentration-dependent relaxant effect for spastic contractions of K^+ (80 mM) and K^+ (25 mM) in jejunum preparation. The respective EC_{50} of Cl.Hexane, Cl.DCM, Cl.Et and Cl.Aq extracts for K^+ (80 mM) were 2.806 mg/ml (95%CI: 2.220 to 3.566 mg/ml), 1.247 mg/ml (95%CI: 0.9639–1.641 mg/ml), 1.178 mg/ml (95%CI: 0.9081–1.549 mg/ml) and 0.2742 mg/ml (95%CI: 0.2172–0.3492 mg/ml) and EC_{50} for K^+ (25 mM) 1.667 mg/ml (95%CI: 1.31–2.164 mg/ml), 0.1894 mg/ml (95%CI: 0.1411–0.2562 mg/ml), 1.252 mg/ml (95%CI: 0.9245–1.736 mg/ml) and 0.3392 mg/ml (95%CI: 0.2484–0.4735 mg/ml) (Fig. 7A). The calcium channel antagonism activity of extracts was validated by building calcium CRCs in previously cytosolic calcium-free jejunum preparations in the absence and presence of *C. lanatus seeds* extracts. At 0.3 and 1 mg/ml, the aqueous extract of *C. lanatus* seeds elicited rightward shifting with suppression of calcium CRCs and had a substantial suppression of contractile response calcium CRCs at 1 mg/ml (Fig. 5). Calcium CRCs were neither inhibited nor partly repressed by n-hexane, dichloromethane, or ethanol extracts of *C. lanatus* seeds, although they induced a rightward shift of calcium CRCs. In the presence of atropine (1 μM), dichloromethane and ethanol extracts suppressed the CRCs of calcium. It suggests the contractile response in the absence of atropine was mediated through cholinomimetic components (Fig. 7B). Atropine did not have any inhibitory effect on CRCs of calcium. Hence, calcium channel blocker (CCB) compounds are prominent in the ethanol and aqueous extracts of *C. lanatus* seeds but less n-hexane and dichloromethane extracts. These findings were confirmed and compared to verapamil, a calcium channel blocker (CCB), to examine whether *C. lanatus* seeds extracts had any calcium antagonistic effect. Verapamil repressed the spontaneous, K^+ (80 mM) and K^+ (25 mM) elicited contractions with respective EC_{50} of 0.05,876 μM (95% Cl:0.04740–0.07,326 μM), 0.1165 μM (95%CI: 0.1003–0.1355 μM) and 0.01,595 μM (95%CI: 0.01303–0.01,956 μM), respectively (Fig. 7A). Additionally, CRCs of calcium were constructed and caused a complete blockade at a dose (0.3 μM) similar to *C. lanatus* seeds extracts (Fig. 7B). Hence it is indicated that antagonism activity of *C. lanatus* seeds extracts towards the calcium ions channels.

Effect on isolated rat ileum preparations

A bolus dose of Cl.DCM and Cl.Et extracts of *C. lanatus* seeds were induced to isolated rat ileum preparations for spasmogenic effect. Both extracts exerted the contractile response at 0.3, 1 and 3 mg/ml in similar manner to acetylcholine (0.1, 0.3 and 1 μM). However, in the presence of atropine (1 μM), the contractile response extracts were significantly ($p < 0.001$ vs. Ach 1 μM) diminished of contraction at low doses (0.3, 1

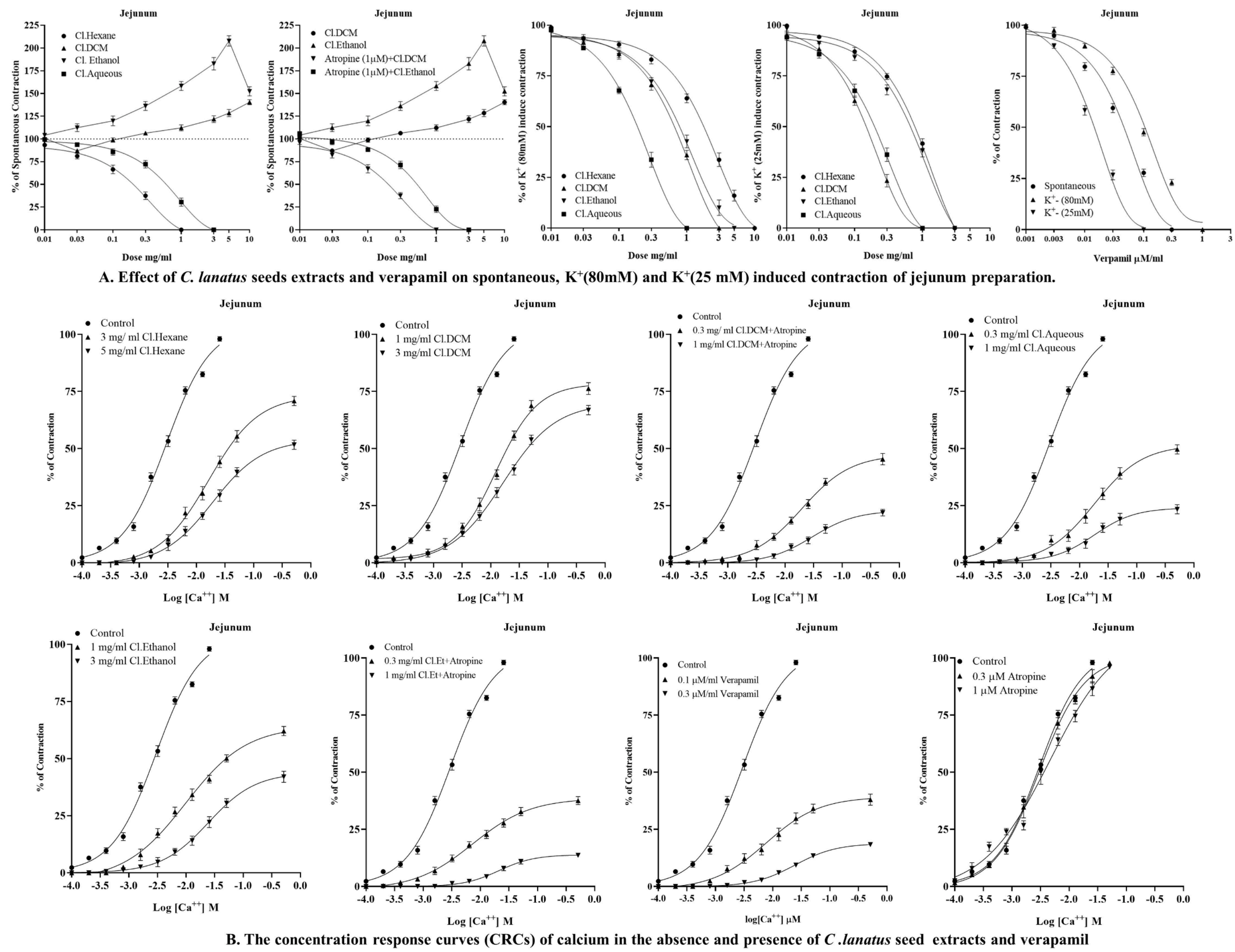


Fig. 7. **(A.)** Effect of sequential extracts of *C. lanatus* seeds and verapamil on jejunum preparation, in respect of spontaneous K^+ (80 mM), carbachol (1 μM), and K^+ (25 mM) induce contraction. **(B.)** The concentration response curves of calcium in presence and absence of sequential extracts of *C. lanatus* seeds and verapamil on jejunum preparation (Values are expressed as Mean ± SD, n = 4, data was analyzed by dose-response curve).

and 3 mg/ml) in a similar manner to acetylcholine contraction response in the presence of atropine (1 μM) (Fig. 8).

Effect on isolated rabbit tracheal preparations

To investigate the possibility of a bronchodilator action, polarity-based sequential extracts of *C. lanatus* seeds were introduced into rabbit tracheal preparations.When subjected to persistent contractions of K^+ (80 mM), CCh (1 μM), and K^+ (25 mM), the Cl.Hexane, Cl.DCM, Cl. Et, and Cl.Aq had a concentration-dependent relaxant effect (Fig. 9A). The Cl.Hexane, Cl.DCM, Cl.Et and Cl.Aq extracts exhibited concentration-dependent inhibitory response for K^+ (80 mM) evoked contractions with respective EC_{50} 12.33 mg/ml (95%CI: 8.747–18.62 mg/ml), 1.055 mg/ml (95%CI: 0.7994–1.417 mg/ml), 5.331 mg/ml (95%CI: 3.382–9.622 mg/ml) and 0.6060 mg/ml (95%CI: 0.4150–0.9474 mg/ml). The respective EC_{50} value of Cl.Hexane, Cl. DCM, Cl.Et and Cl.Aq for CCh (1 μM) induced contractions were 2.134 mg/ml (95%CI: 1.574–3.023 mg/ml), 2.036 mg/ml (95%CI: 1.590–2.659 mg/ml), 3.496 mg/ml (95%CI: 2.629–4.820 mg/ml) and 0.1972 mg/ml (95%CI: 0.1556–0.2510 mg/ml) and for K^+ (25 mM) induced contractions were 0.6118 mg/ml (95%CI: 0.4939–0.7625 mg/ml), 0.5501 mg/ml (95%CI: 0.4108–0.7627 mg/ml), 0.3710 mg/ml (95%CI: 0.2762–0.5095 mg/ml) and 0.2508 mg/ml (95%CI: 0.2056–0.3074 mg/ml). Carbachol CRCs were formed in the presence and absence of Cl.Hexane, Cl.DCM, Cl.Et, and Cl.Aq (1 and 3 mg/ml), and these extracts induced a noncompetitive shift to the right with carbachol CRC suppression (**Fig. 9B**). These findings were confirmed and compared to verapamil, which induced a concentration-dependent inhibitory response to spastic contractions of CCh(1 μM), K^+(80 mM) and K^+(25 mM) in tracheal preparation respective EC_{50} values of 0.1864 μM (95%CI: 0.1259–0.2945 μM), 0.1212 μM (95%CI: 0.1052–0.1397 μM) and 0.01,037 μM (95%CI: 0.008213–0.01,307 μM) (Fig. 9A). CCh CRCs were suppressed by verapamil at dose(1 μM) in a manner similar to that of C. lanatus seeds extracts (Fig. 9B).

Effect on isolated rabbit urinary bladder preparations

The antispasmodic effect of smooth muscles was investigated using polarity-based sequential extracts of *C. lanatus* seeds in rabbit urinary bladder preparations. When tested on CCh (1 M), K^+ (80 mM), and K^+ (25 mM), the Cl.Hexane, Cl.DCM, Cl.Et, and Cl.Aq elicited contractions in a concentration-dependent manner (Fig. 10A). The respective EC_{50} of Cl.Hexane, Cl.DCM, Cl.Et and Cl.Aq extracts for K^+ (80 mM) evoked contractions were 19.21 mg/ml (95%CI: 12.76–33.28 mg/ml), 0.5684 mg/ml (95%CI: 0.4465–0.7288 mg/ml), 1.607 mg/ml (95%CI: 1.293–2.022) and 0.2199 mg/ml (95%CI: 0.1667–0.2924 mg/ml) for CCh (1 μM) persistent contraction were 1.587 mg/ml (95%CI: 1.291–1.980 mg/ml), 0.2540 mg/ml (95%CI: 0.1859–0.3521 mg/ml),

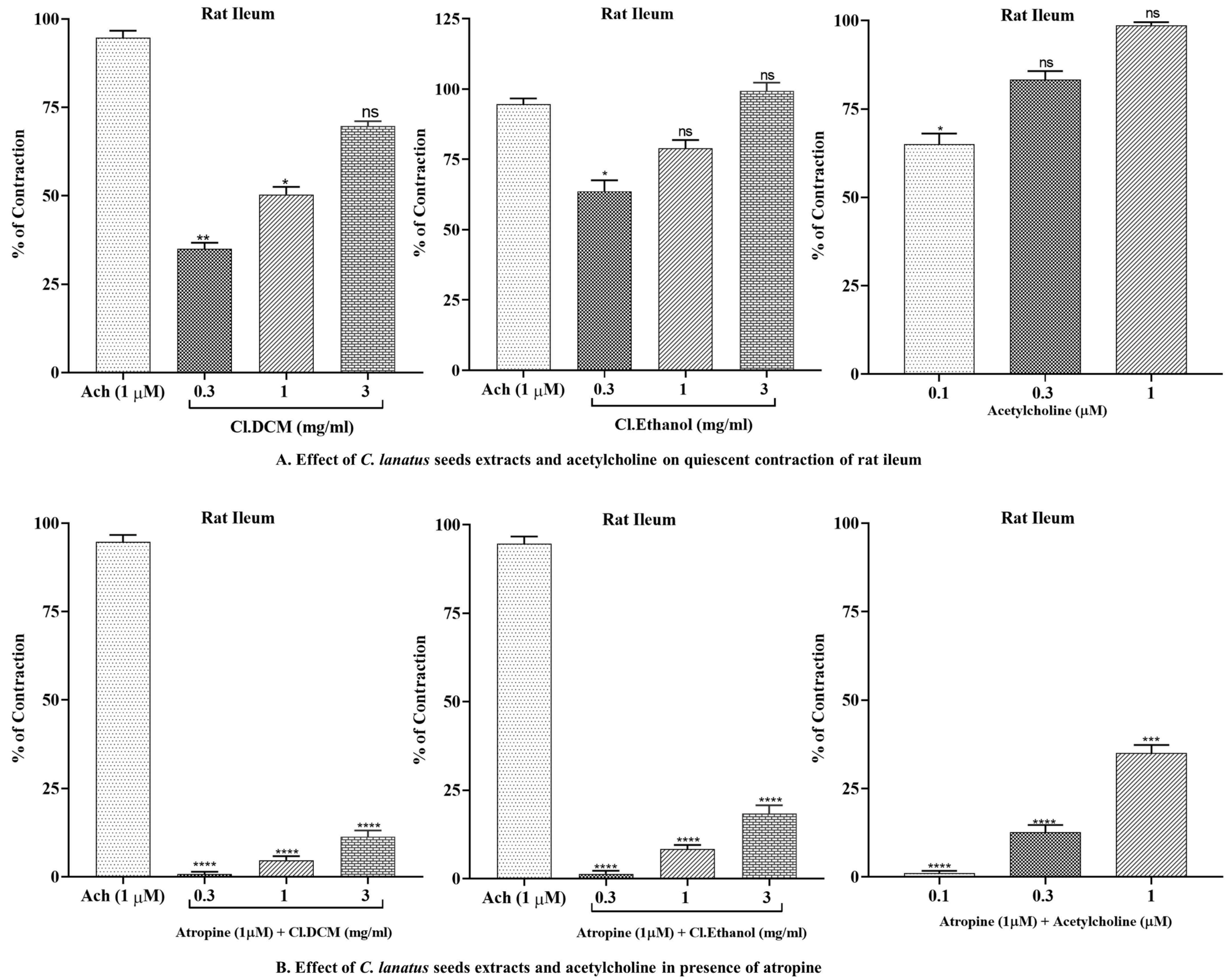


Fig. 8. Effect of dichloromethane and ethanol extracts of *C. lanatus* seeds and acetylcholine on rat ileum preparations in presence and absence of atropine. (Values are expressed as Mean ± SD, n = 4, data was analyzed by one-way ANOVA followed by Dunnett's test when compared to Ach, and $p < 0.05$ was considered significant. (*$p < 0.05$, **$p < 0.01$, *** $p < 0.001$).

0.6436 mg/ml (95%CI: 0.5133–0.8126 mg/ml) and 0.07115 mg/ml (95%CI: 0.04493–0.1140 mg/ml), and K^+ (25 mM) evoked contractions were 2.600 mg/ml (95%CI: 1.877–3.751 mg/ml), 0.1160 mg/ml (95% CI: 0.08073–0.1662 mg/ml), 0.4865 mg/ml (95%CI: 0.3742–0.6483 mg/ml) and 0.07396 mg/ml (95%CI: 0.04527 to 0.1231 mg/ml). These findings were confirmed and compared to verapamil, which induced a concentration-dependent inhibitory response to spastic contractions of CCh(1 μM), K^+(80 mM) and K^+(25 mM) with respective EC_{50} values of 0.05,390 μM (95%CI: 0.04415–0.06,608 μM), 0.1425 μM (95%CI: 0.1162–0.1750 μM) and 0.01,013 μM (95%CI: 0.007456–0.01,370 μM) (Fig. 10A). The Cl.Aq elicited rightward shifting with suppression of calcium CRCs at 1 and 3 mg/ml, with substantial repression of contractile response of calcium CRCs at 3 mg/ml (Fig. 10B). The Cl.Hexane, Cl.DCM, and Cl.Et extracts were partially suppress calcium CRCs but cause rightward shift of calcium CRCs. In the presence of atropine (1 μM), Cl.DCM and Cl.Et extracts suppressed the CRCs of calcium.

In vivo experiments

Maximum tolerated dose

When polarity-based sequential extracts of *C. lanatus* were studied for a maximum tolerated extract dose in rats. It was found safe with no changes in the body weight, clinical signs of distress, behavioral changes, and mortality of rats were observed during 28 days.

Effect on GI charcoal meal intestinal transit

In propulsive movement experiments, pretreatment with Cl.Hexane and Cl.Aq extracts resulted in a dose-dependent (150 and 300 mg/kg) significant reduction ($p < 0.001$ vs. negative control), in charcoal meal propulsive movement whereas Cl.DCM and Cl.Et extracts had no effect on peristaltic movements. Cl.Hexane extract had a peristalsis index of 51.6% at 150 mg/kg and 28% at 300 mg/kg, however Cl.Aq extract had a peristalsis index of 25.5% at 150 mg/kg and 15% at 300 mg/kg, comparable to positive controls loperamide (10 mg/kg) and verapamil (10 mg/kg). At a dose of 300 mg/kg, the peristalsis index of Cl.DCM and Cl.Et extracts was 55.6% and 53.3%, respectively (Fig. 11).

Effect on castor oil-induced diarrhea

Pretreatment of Cl.Hexane and Cl.Aq extracts with castor oil provoked diarrhea showed a dose-dependent (150 and 300 mg/kg) significant ($p < 0.001$ vs. negative control) inhibitory effect, whereas Cl.DCM and Cl.Et extracts did not demonstrate protection against defecation. Cl.

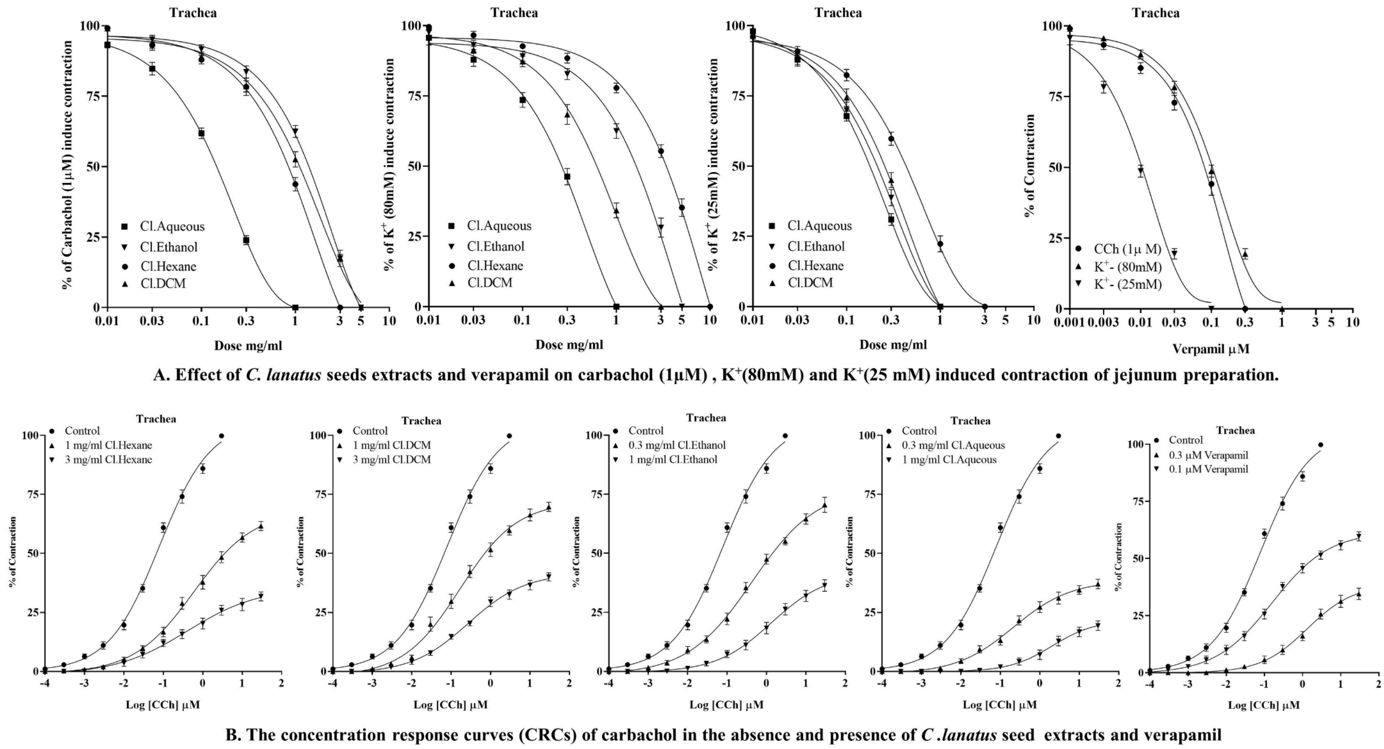


Fig. 9. **(A.)** Effect of sequential extracts of *C. lanatus* seeds and verapamil on tracheal preparation, in respect of K^+ (80 mM), carbachol (1 μM), and K^+ (25 mM) induce contraction. **(B.)** The concentration-response curves of carbachol in presence and absence of sequential extracts of *C. lanatus* seeds and verapamil on jejunum preparation (Values are expressed as Mean ± SD, n = 4, data was analyzed by dose-response curve).

Hexane extract protected mice from castor oil-induced diarrhea 48.7% at 150 mg/kg and 68.3% at 300 mg/kg, although Cl.Aq extract protected animals from castor oil-induced diarrhea 63.6% at 150 mg/kg and 79.6% at 300 mg/kg. Both extracts, like positive controls loperamide (10 mg/kg) and verapamil (10 mg/kg), provide significant protection ($p < 0.001$ vs. negative control) against diarrhea. However, Cl.DCM and Cl. Et extracts failed to inhibit diarrhea—the protection for Cl.DCM and Cl. Et extracts were 22.4% and 42% at dose 300 mg/kg, respectively, non-significant protection from diarrhea (Fig. 11).

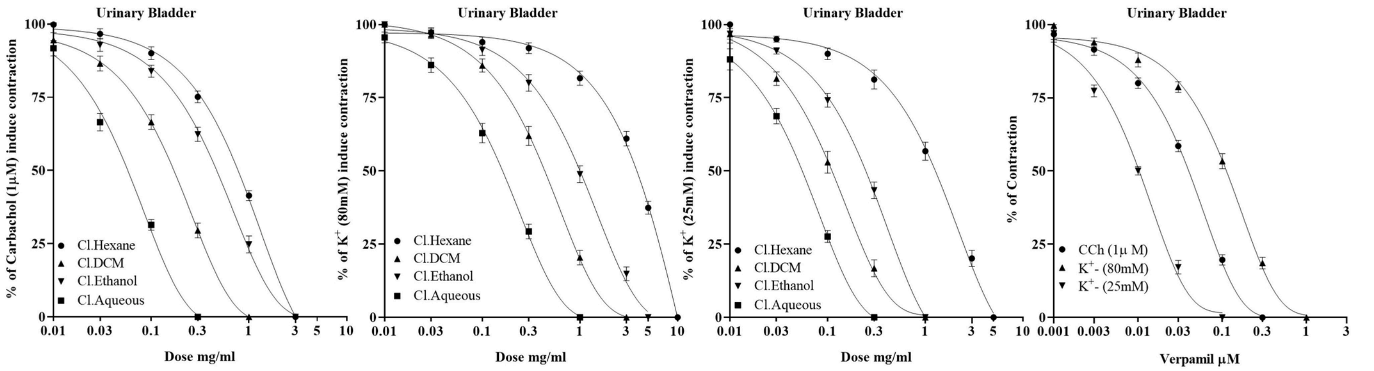
Effect on intestinal fluid accumulation

The oral administration of castor oil resulted in a substantial increase ($p < 0.001$) in fluid accumulation (137 ± 1.7 g) as compared to the control group receiving normal saline (112 ± 1.8 g). Cl.Aq pretreatment elicited a dose-dependent antisecretory response (150 and 300 mg/kg), whereas Cl.Hexane and Cl.Et elicited a moderate antisecretory response. In contrast, Cl.DCM failed to show antisecretory response against castor oil-induced fluid accumulation. Cl.Aq extract significantly ($p < 0.001$ vs castor oil group) reduced the intestine secretions, and the weight of fluid was 97.6 ± 1.5 g at 150 mg/kg and 85.3 ± 0.60 g at dose 300 mg/kg, like positive controls loperamide (76.5 ± 1.4 g) and verapamil (79.0 ± 0.14 g). The fluid accumulation for Cl.Hexane and Cl.Et extracts were 104 ± 1.6 g and 97.3 ± 0.41 g at dose 150 mg/kg and 9.7 ± 1.3 g and 105.6 ± 2.3 g at dose 300 mg/kg, respectively. The fluid weight for Cl.DCM extract was 101.6 ± 2.5 g at 150 mg/kg but increased in fluid weight up to 115.3 ± 1.6 g at dose 300 mg/kg (Fig. 11).

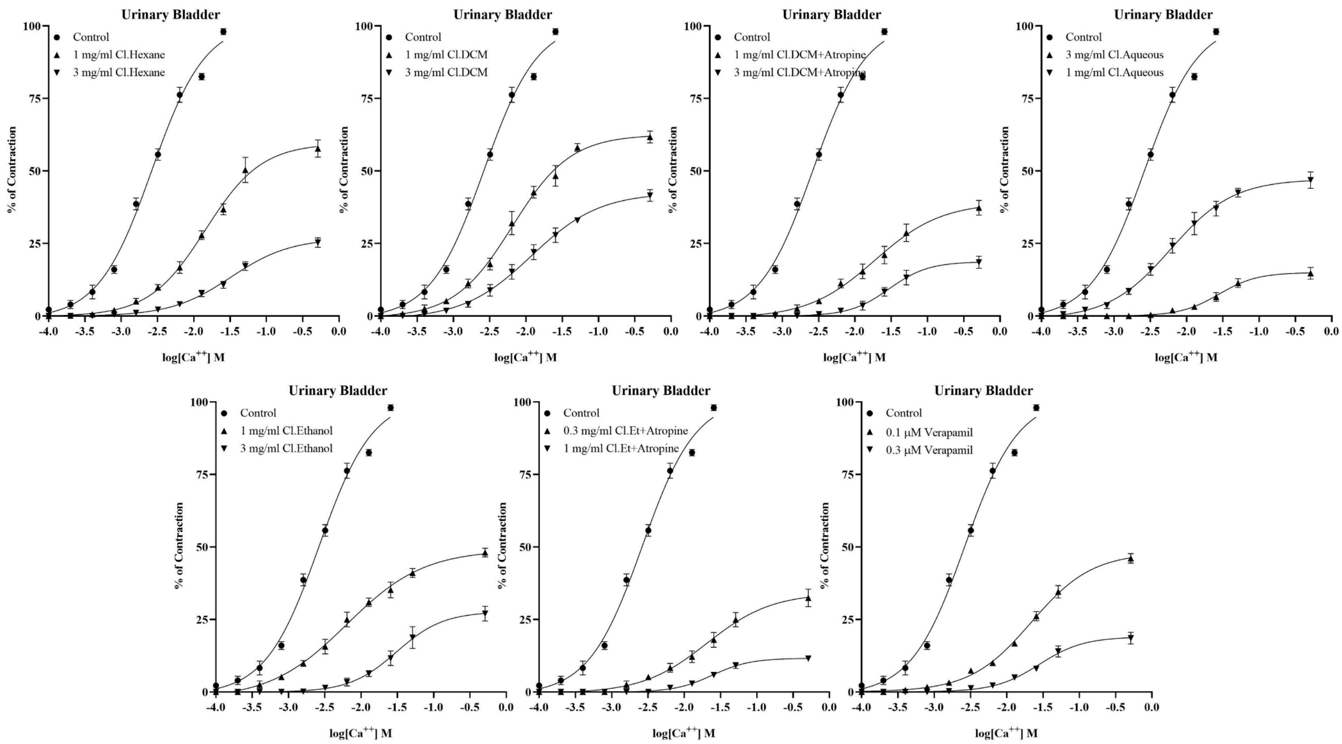
Discussion

Since ancient times, herbal therapy has been widely acknowledged as a safe and effective form of supplementary medicine or dietary patterns for treating a broad variety of chronic illnesses. Traditional healers utilize *C. lanatus* seeds and fruits to treat various diseases (Siddiqui et al., 2018). *Citrullus lanatus* (Thunb.) Matsum. & Nakai. seeds and fruit have been recognized as traditional healers and used to heal multiple ailments. However, more investigation is needed to understand the antispasmodic properties of *C. lanatus* seeds. Therefore, this study was designed to explore potential mechanisms of *C. lanatus* seeds in digestive, respiratory, and urinary by an integrated strategy of KEGG and GO profiling of *C. lanatus* bioactive compounds. LC ESI-MS/MS determined that the sequential extracts of *C. lanatus* seeds contained anthraquinones, sterols, phenols, flavones, and flavonoids. In hexane extract, it contains palmitoleic acid, linoleic acid, stearic acid, oleic acid, and campesterol; in DCM extract, umbelliferone, stigmasterol, and caffeic acid; in ethanol extract, quinic acid, ferulic acid, quercetin, scopoletin, epicatechin, 1,4-dicaffeoylquinic acid, rutin, and apigenin; and in aqueous extract luteolin 7-O-glucoside, p-coumaric acid, malic acid, hesperidin, apigenin-7-O-glucuronide, and kaempferol. The rutin, kaempferol, and quercetin exert antispasmodic activity. These compounds may have an antispasmodic effect (Amira et al., 2008; Gilani et al., 2008).

In 2007, Hopkins (2007) argued that by evaluating drug or ligand cellular processes, network pharmacology elucidates the link between bioactive ingredients, various targets, and pathways of complex diseases (Zhang et al., 2013). Network pharmacology has recently evolved as a new tool for identifying cellular pathways and emphasizing the active compounds of herbs (Park et al., 2018). In computational investigations, bioactive compounds were predicted to interfere with cholinergic synapses and calcium mediated signaling for antispasmodic action. According to KEGG and GO pathway analyzes, the bioactive compounds interacted with target genes involved in smooth muscle contraction and calcium-mediated signaling. Like verapamil, kaempferol, rutin, and quercetin exhibited an apparent binding affinity for L-type voltage-gated calcium ion channels, myosin light chain kinase, and calcium/calmodulin kinase in molecular docking. Consequently, the significant antispasmodic activity of *C. lanatus* seeds may be related to the high binding affinity of compounds for target proteins, which blocks the contraction-causing signal transduction (Figs. 6, and S3, S4).



A. Effect of *C. lanatus* seeds extracts and verapamil on carbachol (1μM) , K⁺(80mM) and K⁺(25 mM) induced contraction of jejunum preparation.



B. The concentration response curves (CRCs) of carbachol in the absence and presence of *C .lanatus* seed extracts and verapamil

Fig. 10. **(A.)** Effect of sequential extracts of *C. lanatus* seeds and verapamil on urinary bladder preparation, in respect of K^+ (80 mM), carbachol (1 μM), and K^+ (25 mM) induce contraction. **(B.)** The concentration-response curves of calcium in presence and absence of sequential extracts of *C. lanatus* seeds and verapamil on jejunum preparation (Values are expressed as Mean ± SD, n = 4, data was analyzed by dose-response curve).

C. lanatus seed extracts were materialized to spontaneous contractions of isolated jejunum, tracheal, and urinary tissue preparation for antidiarrheal and bronchodilator response and diversity in results was found. The n-hexane and aqueous extracts reduced spontaneous jejunum contractions, whereas the dichloromethane and ethanol extracts of *C. lanatus* seed revealed a dose-dependent contractile response. Numerous physiological mediators mediate the stimulatory effects in the gut by increasing cytosolic calcium levels, either through extracellular calcium influx or release from cytosolic stores to elicit depolarization of the resting action potential of tissue preparations. These physiological mediators mediate the stimulatory effects in the gut. Increases in cytosolic calcium levels, whether from external calcium influx or calcium release from cytosolic endoplasmic reticulum storage, elicit depolarization of the resting action potential of jejunum preparations and spontaneous contractions. In hyperactive gastrointestinal illnesses, any drug that can inhibit this mechanism is considered an active treatment agent (Saqib and Janbaz, 2016; Wallace and Sharkey, 2018) and a known antispasmodic agent (Saqib and Janbaz, 2021). Atropine (1 μM) inhibited the spasmogenic response of dichloromethane and ethanol extracts, indicating the presence of cholinomimetic constituents (Gilani et al., 2008).

On jejunum preparations, the spasmolytic ability of *C. lanatus* seed extracts was tested against K^+ (80 mM) and K^+ (25 mM) spastic contractions to see whether it intervenes through calcium channel blockage or potassium channel opening (Gilani et al., 2008). In jejunum preparations, K^+ (80 mM) triggered depolarization with calcium influx into the cell, increasing cytosolic calcium ions, and this intense depolarized membrane action potential led to a sustained contractile response (Saqib and Janbaz, 2016). The smooth muscle will relax after repolarization has occurred. By blocking the influx of calcium current, the polarity-base *C. lanatus* extracts relaxed the K^+ (80 mM) induced contraction and blocked the evoked depolarization and repolarized membrane potential. The following series of events were expected to be inhibited: (1) decreased in cytosolic calcium ions concentration, (2) the insufficient cytosolic calcium ions to interact with the regulating protein phosphokinase C, the calcium-calmodulin complex could not form., (3); myosin light chain kinase (MLCK) activation and MLCs phosphorylation did not occur and (4) The interaction of actin and phosphorylated MLCs

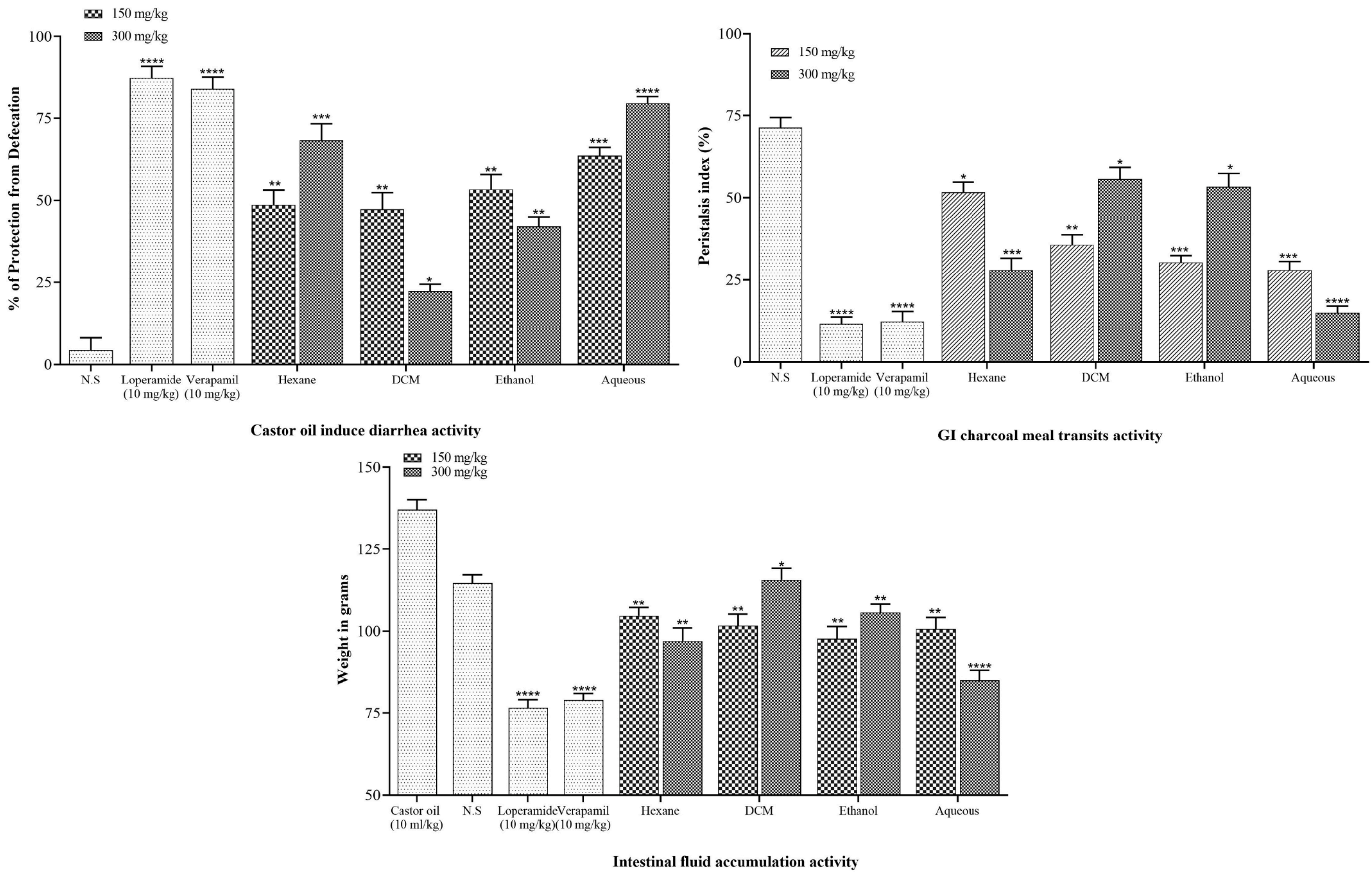


Fig. 11. *In vivo* studies of sequential extracts; antidiarrheal, antiperistalsis, and GI fluid accumulation activities. (Values are expressed as Mean ± SD, n = 5, data was analyzed by one-way ANOVA followed by Dunnett's test for when compared to control (normal saline or castor oil group) and $p < 0.05$ was considered significant. ($*p < 0.05$, $**p < 0.01$, $***\ p < 0.001$).

did not occur; As a result, the tissue contractile response was not attained (Saqib and Janbaz, 2016; Wahid et al., 2021).

The aqueous extract repolarized the K^+ (80 mM) induced depolarization at 1 mg/ml and found more potential CCB among all extracts. The n-hexane (2.806 mg/ml) extract was found with the highest EC_{50} (15.00 mg/ml). Hence it has the lowest CCB activity, while ethanol and DCM extracts had intermediate CCB activity. The pretreatment of ethanol and aqueous extracts caused suppression and a shift to the right of CRCs of calcium on jejunum preparation, similarly to verapamil, which may confirm assumptions that ethanolic and aqueous extracts behaved as CCB and triggered membrane action potential repolarization (Saqib and Janbaz, 2016). However, n-hexane and dichloromethane failed to suppress calcium CRCs. In the presence of atropine (1 μM), both DCM and ethanol extracts showed more potential suppression of CRCs with a rightward shift. It confirmed the speculations that both extracts contain some cholinomimetic and calcium channel blocker constituents (Gilani et al., 2008). In the absence of atropine, cholinomimetic constituents stimulate the muscarinic receptors, which oppose the suppression of calcium CRCs, but in atropine presence, muscarinic receptors were occupied, and cholinergic constituents of extracts failed to exert their cholinomimetic effect (Gilani et al., 2008). K^+ channel openers are drugs that relax the K^+ (25 mM) induced contractions, although calcium channel blockers have an equivalent tendency to inhibit K^+ (80 mM), and K^+ (25 mM) induce contractions. The agents that relaxed the K^+ (25 mM) induce contractions are considered as K^+ channel openers, but calcium channels blockers have an equal tendency to inhibit K^+ (80 mM), and K^+ (25 mM) induce contractions (Saqib and Janbaz, 2021). So, *C. lanatus* seed extracts and verapamil completely blocked K^+ (80 mM), and K^+ (25 mM) induce contractions (Saqib and Janbaz, 2016). As demonstrated in computational studies, kaempferol, rutin, and quercetin have inhibitory effects on the calcium-mediated signaling pathway and alter the cytosolic calcium concentration.

The spasmogenic effect of dichloromethane and ethanol extracts was studied on rat ileum, and it is considered beneficial for measuring spasmogenic response because it acts as quiescent preparation (Gilani et al., 2008; Saqib and Janbaz, 2016). When bolus doses of dichloromethane and ethanol were applied, ileum preparation caused a dose-dependent spasmogenic response like acetylcholine. However, in the presence of atropine, the contractile response was diminished, indicating the presence of cholinomimetic compounds that initiate the stimulation of M_3 muscarinic receptors, but atropine blocked the receptor; these compounds failed to exert their response. Acetylcholine is a cholinergic neurotransmitter that acts on the M_3 muscarinic receptor and plays an influential role in regulating gut peristalsis movements and releasing digestive juices. Atropine is an anticholinergic drug that binds to the M_3 muscarinic receptor and inhibits the cholinergic responses in the gut (Gilani et al., 2008).

The antispasmodic activities of *C. lanatus* extracts were explored by causing depolarization of the action potential tracheal and urinary bladder tissue preparation with carbachol (1 μM) and K^+(80 mM)-induced contractions (Janbaz et al., 2013). All extracts of *C. lanatus* seed exhibited a concentration-dependent inhibitory response for carbachol (1 μM) evoked contraction by maintaining a repolarization state. However, *C. lanatus* seeds showed similar results for K^+(80 mM)-evoked contractions as found in jejunum preparation, i.e., ethanol and aqueous relaxed the contractions. In contrast, dichloromethane extract relaxed the contraction in higher concentrations (10 mg/ml), and n-hexane extract failed to achieve complete relaxation. The EC_{50} value of carbachol (1 μM) indicated that *C. lanatus* seed extracts have muscarinic M_3 receptor activity. The phosphoinositide phospholipase C enzyme is activated when the M_3 muscarinic receptor is activated, and it hydrolyzes phosphatidylinositol 4,5-bisphosphate into two secondary messengers: inositol 1,4,5-triphosphate (IP3) and diacylglycerol (DAG). IP3 stimulates the sarcoplasmic reticulum's inositol 1,4,5-trisphosphate receptors (IP3R) to release calcium ions, increasing cytosolic calcium ion levels. Phosphokinase C (PKC), a regulatory protein activated by diacylglycerol and calcium ions, phosphorylates calmodulin and forms a calcium-calmodulin complex. This complex promotes MLCK activation, resulting in the phosphorylation of MLCs. The phosphorylated MLCs and actin form an interaction network to provide a contractile response (Saqib and Janbaz, 2016). The anticholinergic activity of *C. lanatus* seed extracts was confirmed by a rightward shift in carbachol CRCs on tracheal preparations such as verapamil (Janbaz et al., 2015). As a result, *C. lanatus* extracts decreased cytosolic calcium release from the sarcoplasmic reticulum and blocked the signal transduction of the muscarinic receptor pathway of contractile response, resulting in bronchodilation.

Furthermore, calcium CRCs were built, n-hexane and aqueous extracts repressed calcium CRCs with a shift to the right on urinary bladder preparation, similar to verapamil. At the same time, ethanol suppressed 40–50%, but in the presence of atropine, it suppressed calcium CRCs completely as mentioned *in-silico* results of ethanol, it had dual activity, i.e., calcium mediates signaling process, neurotransmitter levels, and cholinergic synapse. The predicted cholinergic effect was studied on rat ileum preparation in the presence of atropine to obliterate the cholinergic response of dichloromethane and ethanol extracts like acetylcholine (Saqib and Janbaz, 2016).

The aqueous extract exhibited a significant ($p < 0.05$ vs. control) inhibition inducing spasmolytic properties, but dichloromethane and ethanol extracts failed to inhibit peristalsis, diarrhea, and electrolyte imbalance at 150 and 300 mg/kg. The n-hexane extract showed the moderate inhibition of peristalsis, diarrhea, and electrolyte imbalance at 150 and 300 mg/kg. The aqueous extract induced the antispasmodic effect to block the distance traveled by charcoal meal. It was found that aqueous extract blockade the rhythmic contractions of the gut. The hydrolysis of castor oil in the gut produce an active constituent ricinoleic acid (Atta and Mouneir, 2004; Iwao and Terada, 1962) that may increase intestinal fluid contents and promotes electrolyte and water transport imbalance (Gaginella and Phillips, 1975) in the gut, which produces the massive contraction in the transverse and distal colon. The aqueous extract blocked the castor oil-induced diarrhea and fluid secretion. The antiperistalsis, antisecretory and antidiarrheal activities of aqueous extract were mediated through the involvement of calcium channel blocked activity similarly to loperamide and verapamil (Crowe and Wong, 2003; Reynolds et al., 1984). Hence, *C. lanatus* seed extract exerted their bronchodilator and antidiarrheal potential by intervening with cytosolic calcium concentration to inhibit the contractile response. The dichloromethane and ethanol extracts failed to show a significant inhibition against peristaltic movements, castor oil-induced diarrhea and fluid secretion. The dichloromethane and ethanolic extracts contain cholinomimetic phytoconstituents that may increase the gut motility of animals and cause spasms. It indicates the presence of some spasms constituents that cause the failure of extracts to cause inhibition.

.

Conclusion and future perspective

Beside of their nutritional value, *Citrullus lanatus* (Thunb.) seed extracts have therapeutic benefits, which might be related to their relevance in treating asthma, dysuria, and diarrhea. Calcium-mediated signaling to the repolarize membrane action potential modulates the contractile response. In *in-silico, in vitro*, and *in vivo* models, extracts of *C. lanatus* seeds revealed considerable antispasmodic, antiperistalsis, antidiarrheal, dysuria, and anti-asthmatic effects substantiating its traditional claims in Pakistani and Indian traditional medicine systems. Furthermore, the presence of rutin, quercetin, kaempferol, and scopoletin in *C. lanatus* seeds extracts might be regarded as interesting bioactive substances. More research is needed to explain and validate the various modes of action and pathways responsible for the extract's antispasmodic, antiperistalsis, antidiarrheal, antidysuria, dysuria, and anti-asthmatic properties. Finally, our findings imply that extract has antispasmodic, antiperistalsis, antidiarrheal, dysuria, and anti-asthmatic properties, and they give preliminary scientific evidence for the traditional use of *C. lanatus* seeds as a cure for asthma, diarrhea, and dysuria.

Ethical approval bond

Ethical approval was attained from the Ethical Committee of Bahauddin Zakariya University, Multan (EC /04PhDL-S2018) dated 26th March 2018. Researchers agreed using the approved informed consent documented before their enrolment into the study

Funding

This research was partially funded by research grant ORIC/2021/137 awarded by ORIC Bahauddin Zakariya University, Multan to Dr. Fatima Saqib.

CRediT authorship contribution statement

Muqeet Wahid: Conceptualization, Investigation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Fatima Saqib:** Conceptualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

Authors announce that no clash of interests concerning the publication of this research article.

Acknowledgments

This article is part of the Ph.D. thesis of Mr. Muqeet Wahid carried under the supervision of Dr. Fatima Saqib. The author (s) are very thankful to folk healers from Punjab Province for sharing their knowledge regarding folkloric use. The author(s) are very thankful to the Chairman Department of Pharmacology, Dr. Imran, for the Provision of animal house and lab-related facilities.

References

Abu-Reidah, I.M., Arráez-Román, D., Segura-Carretero, A., Fernández-Gutiérrez, A., 2013. Profiling of phenolic and other polar constituents from hydro-methanolic extract of watermelon (Citrullus lanatus) by means of accurate-mass spectrometry (HPLC-ESI-QTOF-MS). Food Res. Int. 51, 354–362. https://doi.org/10.1016/j.foodres.2012.12.033.

Alka, G., Anamika, S., Ranu, P., 2012. Compositional studies of <italic>Citrullus lanatus</italic>(Egusi melon) seed. Internet J. Nutr. Wellness 6. https://doi.org/10.5580/e6f.

Amira, S., Rotondo, A., Mulè, F., 2008. Relaxant effects of flavonoids on the mouse isolated stomach: structure-activity relationships. Eur. J. Pharmacol. 599, 126–130. https://doi.org/10.1016/j.ejphar.2008.09.021.

Atta, A.H., Mouneir, S.M., 2004. Antidiarrhoeal activity of some Egyptian medicinal plant extracts. J. Ethnopharmacol. 92, 303–309. https://doi.org/10.1016/j.jep.2004.03.017.

Crowe, A., Wong, P., 2003. Potential roles of P-gp and calcium channels in loperamide and diphenoxylate transport. Toxicol. Appl. Pharmacol. 193, 127–137. https://doi.org/10.1016/S0041-008X(03)00372-7.

El-Adawy, T.A., Taha, K.M., 2001. Characteristics and composition of watermelon, pumpkin, and paprika seed oils and flours. J. Agric. Food Chem. 49, 1253–1259. https://doi.org/10.1021/jf001117+.

Elasoru, S.E., Rhana, P., de Oliveira Barreto, T., Naves de Souza, D.L., Menezes-Filho, J.E. R., Souza, D.S., Loes Moreira, M.V., Gomes Campos, M.T., Adedosu, O.T., Roman-Campos, D., Melo, M.M., Cruz, J.S., 2021. Andrographolide protects against

isoproterenol-induced myocardial infarction in rats through inhibition of L-type Ca^{2+} and increase of cardiac transient outward K^+ currents. Eur. J. Pharmacol. 906, 174194.
Erhirhie, E., Ekene, N., 2014. Medicinal values on Citrullus lanatus (Watermelon): pharmacological review. Int. J. Res. Pharm. Biomed. Sci. 4, 1305–1312.
Gaginella, T.S., Phillips, S.F., 1975. Ricinoleic acid: current view of an ancient oil. Am. J. Dig. Dis. 20, 1171–1177. https://doi.org/10.1007/BF01070759.
Gilani, A.H., Khan, A.U., Raoof, M., Ghayur, M.N., Siddiqui, B.S., Vohra, W., Begum, S., 2008. Gastrointestinal, selective airways and urinary bladder relaxant effects of Hyoscyamus niger are mediated through dual blockade of muscarinic receptors and Ca^{2+} channels. Fundam. Clin. Pharmacol. 22, 87–99. https://doi.org/10.1111/j.1472-8206.2007.00561.x.
Gill, N.S., Sood, S., Muthuraman, A., Bali, M., Dev Sharma, P., 2011. Evaluation of antioxidant and anti-ulcerative potential of Citrullus lanatus seed extract in rats. Lat. Am. J. Pharm. 30, 429–434.
Hopkins, A.L., 2007. Network pharmacology. Nat. Biotechnol. 25, 1110–1111. https://doi.org/10.1038/nbt1007-1110.
Iwao, I., Terada, Y., 1962. On the mechanism of diarrhea due to castor oil. Jpn. J. Pharmacol. 12, 137–145. https://doi.org/10.1254/jjp.12.137.
Janbaz, K.H., Nisa, M., Saqib, F., Imran, I., Zia-Ul-Haq, M., De Feo, V., 2013. Bronchodilator, vasodilator and spasmolytic activities of methanolic extract of Myrtus communis L. J. Physiol. Pharmacol. 64, 479–484.
Janbaz, K.H., Zaeem Ahsan, M., Saqib, F., Imran, I., Zia-Ul-Haq, M., Abid Rashid, M., Jaafar, H.Z.E., Moga, M., 2015. Scientific basis for use of Pyrus pashia Buch.-Ham. ex D. Don. fruit in gastrointestinal, respiratory and cardiovascular ailments. PLoS One 10, e0118605. https://doi.org/10.1371/journal.pone.0118605.
Khare, C.P., 2007. Indian Medicinal Plants, Indian Medicinal Plants. Springer, New York, New York, NY. https://doi.org/10.1007/978-0-387-70638-2.
Kuhn, B., Kollman, P.A., 2000. Binding of a diverse set of ligands to avidin and streptavidin: an accurate quantitative prediction of their relative affinities by a combination of molecular mechanics and continuum solvent models. J. Med. Chem. 43, 3786–3791. https://doi.org/10.1021/jm000241h.
Mabaleha, M.B., Mitei, Y.C., Yeboah, S.O., 2007. A comparative study of the properties of selected melon seed oils as potential candidates for development into commercial edible vegetable oils. JAOCS J. Am. Oil Chem. Soc. 84, 31–36. https://doi.org/10.1007/s11746-006-1003-7.
Massa, N.M.L., Silva, A.S., Toscano, L.T., Silva, J.D., gomes, R., Persuhn, D.C., Gonçalves, M.D.C.R., 2016. Watermelon extract reduces blood pressure but does not change sympathovagal balance in prehypertensive and hypertensive subjects. Blood Press. 25, 244–248. https://doi.org/10.3109/08037051.2016.1150561.
Morais, D.R., Rotta, E.M., Sargi, S.C., Bonafe, E.G., Suzuki, R.M., Souza, N.E., Matsushita, M., Visentainer, J.V., 2017. Proximate composition, mineral contents and fatty acid composition of the different parts and dried peels of tropical fruits cultivated in Brazil. J. Braz. Chem. Soc. 28, 308–318. https://doi.org/10.5935/0103-5053.20160178.
Ouassor, I., Aqil, Y., Belmaghraoui, W., El Hajjaji, S., 2020. Characterization of two Moroccan watermelon seeds oil varieties by three different extraction methods. OCL Oilseeds Fats Crop. Lipids 27. https://doi.org/10.1051/ocl/2020010.
Park, M., Park, S.Y., Lee, H.J., Kim, C.E., 2018. A systems-level analysis of mechanisms of platycodon grandiflorum based on a network pharmacological approach. Molecules 23, 2841. https://doi.org/10.3390/molecules23112841.
Patel, S., Rauf, A., 2017. Edible seeds from cucurbitaceae family as potential functional foods: immense promises, few concerns. Biomed. Pharmacother. 91, 330–337. https://doi.org/10.1016/j.biopha.2017.04.090.
Perkins-Veazie, P., Collins, J.K., Clevidence, B., Wu, G., 2007. Watermelons and health. Acta Hortic. 121–127. https://doi.org/10.17660/actahortic.2007.731.17.
Qamar, M., Akhtar, S., Ismail, T., Yuan, Y., Ahmad, N., Tawab, A., Ismail, A., Barnard, R. T., Cooper, M.A., Blaskovich, M.A.T., Ziora, Z.M., 2021. Syzygium cumini(L.), skeels fruit extracts: *in vitro* and *in vivo* anti-inflammatory properties. J. Ethnopharmacol. 271, 113805 https://doi.org/10.1016/j.jep.2021.113805.
Rabiu, S., Muhammad, S.A., 2015. Antiulcerogenic effect of aqueous seed and rind extracts of Citrullus lanatus (Water Melon) in ethanol-induced gastric ulcer albino rats. Niger. J. Basic Appl. Sci. 23, 111–115. https://doi.org/10.4314/njbas.v23i2.4.
Raihana, A.R.N., Marikkar, J.M.N., Amin, I., Shuhaimi, M., 2015. A review on food values of selected tropical fruits seeds. Int. J. Food Prop. 18, 2380–2392. https://doi.org/10.1080/10942912.2014.980946.
Reetu, Tomar, M., 2017. Watermelon: a valuable horticultural crop with nutritional benefits. Pop. Kheti 5, 5–9.
Renner, S., Pandey, A., 2013. The cucurbitaceae of India: accepted names, synonyms, geographic distribution, and information on images and DNA sequences. PhytoKeys 20, 53–118. https://doi.org/10.3897/phytokeys.20.3948.
Reynolds, I.J., Gould, R.J., Snyder, S.H., 1984. Loperamide: blockade of calcium channels as a mechanism for antidiarrheal effects. J. Pharmacol. Exp. Ther. 231, 628–632.
Rowan, A.N., 1979. Guide For The Care and Use of Laboratory Animals. National Academies Press, Washington, D.C., U.S.A. https://doi.org/10.1159/000460187. J. Med. Primatol.
Saqib, F., Janbaz, K.H., 2021. Ethnopharmacological basis for folkloric claims of Anagallis arvensis Linn. (Scarlet Pimpernel) as prokinetic, spasmolytic and hypotensive in province of Punjab, Pakistan. J. Ethnopharmacol. 267, 113634 https://doi.org/10.1016/j.jep.2020.113634.
Saqib, F., Janbaz, K.H., 2016. Rationalizing ethnopharmacological uses of Alternanthera sessilis: a folk medicinal plant of Pakistan to manage diarrhea, asthma and hypertension. J. Ethnopharmacol. 182, 110–121. https://doi.org/10.1016/j.jep.2016.02.017.
Seraglio, S.K.T., Valese, A.C., Daguer, H., Bergamo, G., Azevedo, M.S., Gonzaga, L.V., Fett, R., Costa, A.C.O., 2016. Development and validation of a LC-ESI-MS/MS method for the determination of phenolic compounds in honeydew honeys with the diluted-and-shoot approach. Food Res. Int. 87, 60–67. https://doi.org/10.1016/j.foodres.2016.06.019.
Sharkey, K.A., MacNaughton, W.K., 2018. Gastrointestinal motility and water flux, emesis, and biliary and pancreatic disease. In: Brunton, L.L., Hilal-Dandan, R., Knollmann, B.C. (Eds.), Goodman & Gilman's The Pharmacological Basis of Therapeutics. McGraw-Hill, New York, NY, USA, pp. 921–944.
Sharma, S., Yadav, S., Singh, G., Paliwal, S., Dwivedi, J., 2012. First report on laxative activity of Cucumis Sativus. Int. J. Pharm. Sci. Rev. Res. 12, 129–131.
Siddiqui, W.A., Shahzad, M., Shabbir, A., Ahmad, A., 2018. Evaluation of anti-urolithiatic and diuretic activities of watermelon (Citrullus lanatus) using *in vivo* and *in vitro* experiments. Biomed. Pharmacother. 97, 1212–1221. https://doi.org/10.1016/j.biopha.2017.10.162.
Sirous, H., Chemi, G., Campiani, G., Brogi, S., 2019. An integrated *in silico* screening strategy for identifying promising disruptors of p53-MDM2 interaction. Comput. Biol. Chem. 83 https://doi.org/10.1016/j.compbiolchem.2019.107105.
Syed, M., Khan, M.N., Khadim, A., Shadab, H., Perveen, A., El-Seedi, H.R., Musharraf, S. G., 2021. Chemical fingerprinting of three Anemone species and an adulteration study to detect cross mixing of medicinal plants by HPLC-HR-ESI-MS/MS method. J. King Saud Univ. Sci. 33, 101461 https://doi.org/10.1016/j.jksus.2021.101461.
Taiwo, A., Agbotoba, M., Oyedepo, J., Shobo, O., Oluwadare, I., Olawunmi, M., 2009. Effects of drying methods on properties of watermelon (Citrullus lanatus) seed oil. Afr. J. Food Agric. Nutr. Dev. 8 https://doi.org/10.4314/ajfand.v8i4.19208.
Wahid, M., Ali, Anam, Saqib, F., Aleem, A., Bibi, S., Afzal, K., Ali, Atif, Baig, A., Khan, S. A., Bin Asad, M.H.H., 2020. Pharmacological exploration of traditional plants for the treatment of neurodegenerative disorders. Phyther. Res. 34, 3089–3112. https://doi.org/10.1002/ptr.6742.
Wahid, M., Saqib, F., Ahmedah, H.T., Gavris, C.M., De Feo, V., Hogea, M., Moga, M., Chicea, R., 2021. Cucumis sativus L. seeds ameliorate muscular spasm-induced gastrointestinal and respiratory disorders by simultaneously inhibiting calcium mediated signaling pathway. Pharmaceuticals 14, 1197. https://doi.org/10.3390/ph14111197.
Wallace, J.L., Sharkey, K.A., 2018. Pharmacotherapy of gastric acidity, peptic ulcers, and gastroesophageal reflux disease. In: Brunton, L.L., Hilal-Dandan, R., Knollmann, B.C. (Eds.), Goodman & Gilman's The Pharmacological Basis of Therapeutics. McGraw-Hill, New York, NY, USA, pp. 1309–1322.
Xiao, P.ting, Liu, S.yu, Kuang, Y.jia, Jiang, Z.meng, Lin, Y., Xie, Z.shen, Liu, E.H., 2021. Network pharmacology analysis and experimental validation to explore the mechanism of sea buckthorn flavonoids on hyperlipidemia. J. Ethnopharmacol. 264, 113380 https://doi.org/10.1016/j.jep.2020.113380.
Zhang, G.B., Li, Q.Y., Chen, Q.L., Su, S.B., 2013. Network pharmacology: a new approach for Chinese herbal medicine research. Evidence-based complement. Altern. Med. 2013 https://doi.org/10.1155/2013/621423.